

Screw Conveyor Design

Final Project Report

ME 5327 Design for Manufacturing

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Abstract

This project presents the design and evaluation of an industrial screw conveyor system developed for continuous handling of crushed limestone. The system is designed to transport material at a capacity of 150 tons per hour over a horizontal distance of 30 feet while maintaining reliable operation and effective dust containment. The design process begins with the development of a detailed product design specification, followed by functional decomposition and generation of multiple design concepts.

A weighted decision matrix was used to evaluate the concepts based on performance, manufacturability, cost, and reliability, leading to the selection of the most balanced design. The chosen concept was further developed into a complete CAD model, including detailed assembly and component-level designs. Design for Manufacturing (DFM) and Design for Assembly (DFA) analyses were conducted to assess fabrication feasibility and assembly efficiency.

The DFM results indicate that the design is compatible with standard manufacturing processes such as sheet metal forming and machining, with minor opportunities for simplification. The DFA analysis highlights a relatively high assembly complexity due to the number of fasteners and components, suggesting areas for improvement through modularization and part reduction. Overall, the project demonstrates a systematic approach to designing a functional, manufacturable, and industry-relevant screw conveyor system.

Table of Contents

Abstract.....	2
1. Introduction.....	6
2. Concept Design Review.....	7
2.1 House of Quality.....	7
2.1.1. Room 1: Customer Requirements.....	7
2.1.2. Room 2: Engineering Characteristics.....	8
2.1.3. Room 3: Correlation Matrix.....	8
2.1.4. Room 4: Relationship Matrix.....	9
2.1.5. Room 5: Importance Ranking.....	10
2.1.6. Room 6: Customer Assessment of Competing Products.....	10
2.1.7. Room 7: Engineering Assessment of Competing Products.....	11
2.1.8. Room 8: Target Values.....	11
2.2 Product Design Specification.....	12
2.3 Physical Decomposition.....	13
2.4 Functional Analysis.....	14
2.5 Product Architecture.....	15
2.5.1 Housing and Containment.....	15
2.5.2 Conveying and Support.....	15
2.5.3 Drive and Power Transmission.....	15
2.5.4 Protection and Maintenance.....	15
2.6 Morphological Chart.....	16
2.7 Concept Design.....	16
2.7.1 Concept 1.....	17
2.7.2 Concept 2.....	17
2.7.3 Concept 3.....	18
2.7.4 Concept 4.....	18
2.8 Objective Tree.....	19
2.9 Concept Generation and Selection.....	20
3. Preliminary Design Review.....	22
3.1 Parametric Design.....	22
3.1.1 Step 1: Formulation of Design Problem.....	22
3.1.2 Step 2: Generate Alternative Designs.....	24
3.1.3 Step 3: Analyze the Alternative Designs.....	27

3.1.4 Step 4: Evaluate the Results of the Analyses.....	28
3.1.5 Step 5: Refine or Optimize the Design.....	30
3.2 Material Selection.....	30
3.2.1. Ashby Chart Screening.....	30
3.2.2. Material Screening for Major Conveyor Components.....	31
3.2.3. Material Property Comparison.....	32
3.2.4. Weighted Decision Matrix for Screw Flights.....	32
3.2.5. Screw Flights.....	33
3.2.6. Screw Shaft, Drive Shaft, Coupling Shaft, and End Shaft.....	33
3.2.7. Trough.....	33
3.2.8. Covers and Cover with Inlet.....	33
3.2.9. Hanger Bearing.....	34
3.2.10. Final Material Selection.....	34
3.3 Manufacturing Process Selection.....	34
3.3.1. Key Manufacturing Components.....	34
3.3.2. Economic Batch Size Screening.....	35
3.3.3. Shape, Size, and Thickness Screening.....	35
3.3.4. PRIMA Process Selection.....	37
3.3.5. Recommended Manufacturing Processes for Key Components.....	38
3.4 Tolerance Stack-up Analysis.....	38
3.4.1. Tolerance Analysis of Conveyor Length Stack-up (Shaft–Coupling–Bearing Alignment).....	38
3.4.2. Tolerance Analysis of Screw–Trough Clearance.....	41
3.5 DFM Analysis.....	43
3.6 DFA Analysis.....	46
4. Final Design.....	49
4.1 Safety Factor Analysis.....	49
4.2 Reliability Analysis.....	51
4.2.1. Reliability Model and Failure Rates.....	51
4.2.2. Reliability of the Chosen Design.....	52
4.2.3. System Reliability Analysis.....	53
4.2.4. Ways to Improve Reliability.....	54
4.2.5. Mean Time to Failure.....	54
4.2.6. Fault Tree Analysis.....	55

4.3 Failure Modes and Effects Analysis.....	56
4.3.1. FMEA Evaluation and Risk Assessment.....	56
4.4 Cost Analysis.....	57
4.4.1. Material Cost Estimation.....	58
4.4.2. Labor Cost Estimation.....	59
4.4.3. Equipment Cost Estimation.....	59
4.4.3. Overhead Cost Estimation.....	60
4.4.4. Tooling Cost Estimation.....	60
4.4.4. Total Estimated Cost Per Unit.....	60
4.4.5. Cost Analysis Key Insights.....	61
4.5 Ergonomics and Safety Analysis.....	61
4.5.1. Design for Safety Approach.....	62
4.5.2. Main Safety Concerns and Design Controls.....	62
4.5.3. Ergonomic Improvements.....	63
4.5.4. Safety Labels Applied.....	63
4.6 CAD Design and Modeling.....	64
4.6.1. Assembly Overview.....	64
4.6.2. Functional Component Description.....	65
4.7 Component-Level CAD Design.....	65
4.7.1. Design Considerations for Manufacturability.....	66
4.7.2. Design Considerations for Assembly.....	67
5. Conclusion.....	67
References.....	69

1. Introduction

This project is based on an industry-inspired design problem developed from requirements provided by a representative from Martin Sprocket and Gear during a class presentation. The representative presented a realistic engineering scenario and described the functional, performance, and mechanical requirements for a screw conveyor system used in industrial material handling. The objective of the project is to design a system that satisfies these requirements while following a structured engineering design process.

The design focuses on developing a cost-optimized screw conveyor system to handle crushed limestone at a capacity of 150 tons per hour over a horizontal length of 30 ft. The material is free-flowing but exhibits aeration and dusting, which requires the design to ensure continuous and stable flow without clogging or material buildup. The system must also maintain dust-tight operation in a partially indoor environment while minimizing maintenance requirements and limiting the use of hanger bearings.

The project was completed by a team of four graduate-level mechanical engineering students who worked collaboratively to develop the design. The approach integrates mechanical design, material selection, manufacturing considerations, and system-level evaluation with a strong emphasis on reducing cost while maintaining performance and reliability.

This document presents the final design report, which compiles the work completed throughout the project in three main stages. The concept design review section presents the development of the product design specification using quality function deployment, generation of multiple design concepts, and selection of the final concept using a structured decision-making approach. The preliminary design review section focuses on material and manufacturing process selection along with detailed design for manufacturing and design for assembly analyses, as well as tolerance stack up evaluation for critical components. The final design review section presents the evaluation of the completed design, including reliability analysis, cost analysis, safety considerations, failure modes and effects analysis, and the development of the complete three-dimensional CAD model of the system.

Figure 1 presents the overall structure of the project in the form of a project overview framework. It illustrates the progression from concept design to preliminary design and final design, along with the key analyses and deliverables associated with each stage. This framework provides a clear representation of how the different components of the project are organized and integrated within this report.

The report provides detailed results and discussion of all design decisions including conveyor sizing, power requirements, component selection, and engineering analyses. It also includes complete engineering drawings with defined dimensions, tolerances, material specifications, and manufacturing processes to fully describe the final design.

PROJECT OVERVIEW FRAMEWORK

This final design report compiles the work completed in three major stages of the design process.

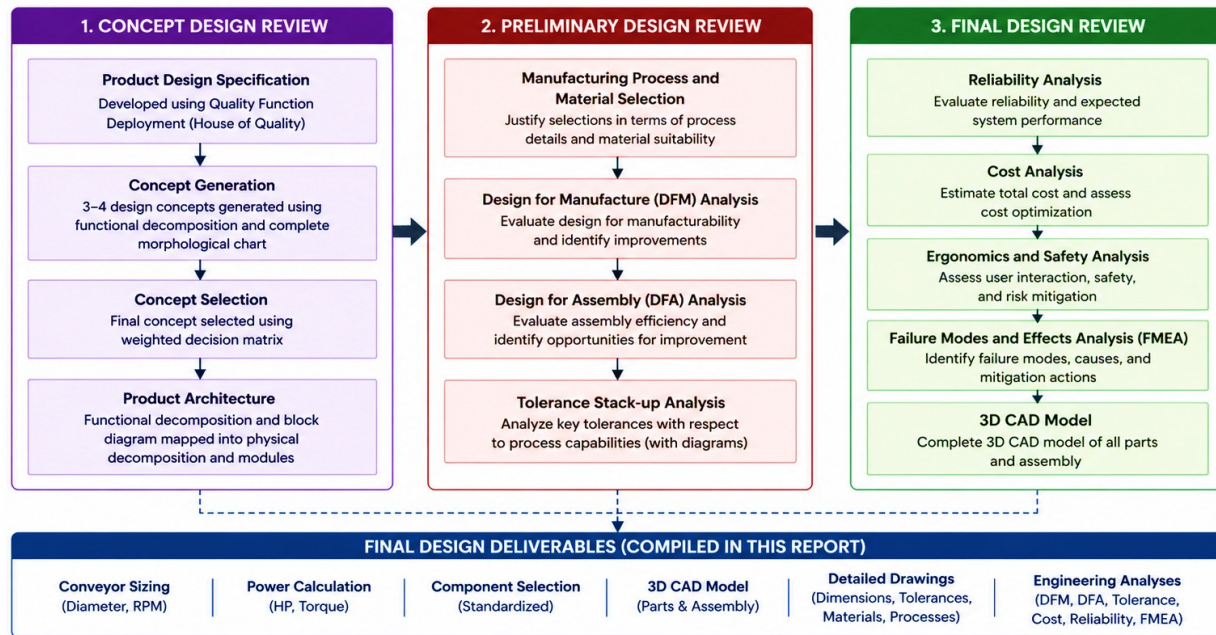


Figure 1. Project Overview Framework Showing the Structure of the Design Process and Report.

2. Concept Design Review

The concept design review establishes the foundation of the screw conveyor design by defining the problem in engineering terms and systematically translating requirements into design decisions. This section begins with the development of the product design specification using quality function deployment, including the construction of the House of Quality to relate customer requirements to engineering characteristics. Based on this framework, multiple design concepts are generated using functional decomposition and a complete morphological approach. A structured evaluation process is then applied to select the final concept using a weighted decision matrix. The selected concept is further refined by defining the product architecture, where functional elements are mapped into physical components and modules that form the basis of the final design.

2.1 House of Quality

The House of Quality has been developed to connect the limestone screw conveyor requirements to measurable engineering characteristics. The conveyor carries crushed limestone at 150 TPH over a 30 ft horizontal distance with a maximum lump size of 2 in. Since the material is dusty, abrasive, and free-flowing, the HOQ focuses on capacity, stable flow, abrasion resistance, dust containment, low maintenance, low assembly complexity, and manufacturing cost. The HOQ is divided into eight rooms: customer requirements, engineering characteristics, correlation matrix, relationship matrix, importance ranking, customer assessment, engineering assessment, and final target values.

2.1.1. Room 1: Customer Requirements

Room 1 lists the main customer requirements for the screw conveyor. High conveying capacity and lump handling receive the highest importance factor because the conveyor has a fixed 150 TPH requirement and accepts 2 in limestone lumps. Stable material flow and wear resistance also rank high because

limestone abrasion and flow interruption directly affect downtime. Dust-tight safe operation, low manufacturing cost, low assembly complexity, and low noise level remain important because the conveyor operates in an industrial environment with maintenance and safety concerns.

Table 2.1. Customer Requirements

Customer Requirements	Preference	Importance Factor
CR1: High conveying capacity	Need	5
CR2: Continuous and stable material flow	Need	4
CR3: Handle lump size and uniform feed	Need	5
CR4: Dust-tight safe operation	Need	3
CR5: High abrasion and wear resistance	Need	4
CR6: Reliable operation and long service life	Want	2
CR7: Low maintenance frequency and downtime	Want	1
CR8: Energy efficient operation	Want	3
CR9: Low manufacturing cost	Need	3
CR10: Low assembly complexity and part count	Need	4
CR11: Low noise level	Need	3
CR12: Minimal obstruction to material flow	Want	1

2.1.2. Room 2: Engineering Characteristics

Room 2 translates the customer requirements into measurable engineering characteristics. These characteristics control final conveyor performance and give the design team specific values to evaluate. Screw diameter, screw speed, throughput capacity, lump size, trough loading, shaft torque, shaft deflection, hanger bearing count, sealing performance, motor power, and manufacturing cost were selected because they directly affect limestone flow, wear, reliability, assembly effort, and cost.

Table 2.2. Engineering Characteristics

Engineering Characteristic	Unit	Direction
EC1: Screw diameter	in	
EC2: Screw rotational speed	RPM	
EC3: Conveyor throughput capacity	TPH	
EC4: Maximum lump size	in	
EC5: Trough loading	CEMA class	
EC6: Enclosure and sealing system performance	Rating	
EC7: Number of intermediate hanger bearings	Count	
EC8: Shaft diameter	in	
EC9: Allowable shaft torque	Torque capacity	
EC10: Shaft deflection	in or mm	
EC11: Installed motor power	HP	
EC12: Manufacturing cost	USD	

2.1.3. Room 3: Correlation Matrix

Room 3 identifies the relationships between engineering characteristics. Increasing screw diameter improves capacity and permits lower screw speed, but it also increases component size, weight, and cost. Lower screw speed reduces wear, dust agitation, noise, and maintenance demand. Reducing the number of hanger bearings improves material flow and lowers internal obstruction, but it increases the importance of shaft sizing and deflection control. Increasing sealing performance improves dust containment and safety, but it adds cover, seal, and fastening requirements. These correlations direct the design away from

the smallest feasible screw and toward the 30 in screw because it balances capacity, speed, cost, and maintenance.

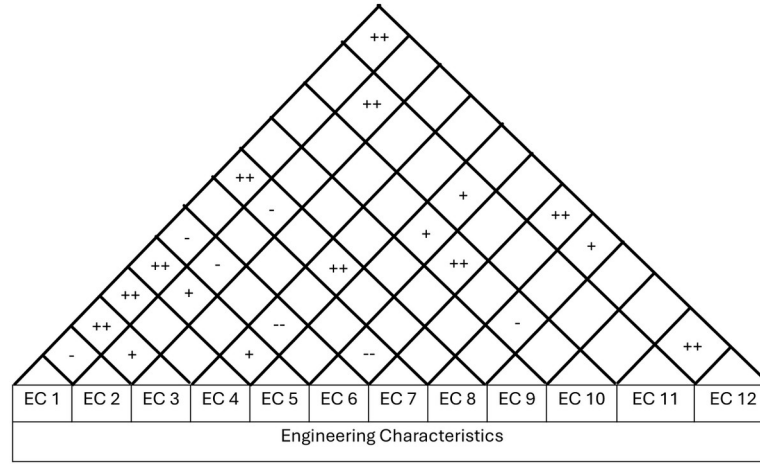


Figure 2. Correlation Matrix.

2.1.4. Room 4: Relationship Matrix

Room 4 connects each customer requirement to the engineering characteristics. Strong relationships are assigned where an engineering characteristic directly controls a customer requirement. High conveying capacity is strongly related to screw diameter, screw speed, throughput capacity, trough loading, and motor power. Dust-tight safe operation is strongly related to enclosure and sealing performance. Low maintenance is strongly related to hanger bearing count, shaft deflection, and sealing performance. The relationship matrix shows that the design is driven mainly by trough loading, screw speed, throughput capacity, screw diameter, motor power, hanger reduction, and sealing.

Table 2.3. Relationship Matrix and Importance Ranking

CR	Imp.	EC1	EC2	EC3	EC4	EC5	EC6	EC7	EC8	EC9	EC10	EC11	EC12
CR1	5	9	9	9		9			3	3	1	9	
CR2	4		9	9	3	9	3						
CR3	5	3			9	3							
CR4	3					3	9	1					
CR5	4	3							3				
CR6	2						3		3	3	9		
CR7	1						3	9			3		
CR8	3					3				9	1	9	
CR9	3						1	1	1				9
CR10	4	1					3	9	1				3
CR11	3		3					3		3	3		
CR12	1				3	9		9					
Raw Score		76	90	81	60	123	63	69	40	57	38	72	39
Relative Weight		9.41	11.14	10.02	7.43	15.22	7.80	8.54	4.95	7.05	4.70	8.91	4.83
Rank		4	2	3	8	1	7	6	10	9	12	5	11

2.1.5. Room 5: Importance Ranking

Room 5 ranks the engineering characteristics using the customer importance factors and relationship values. Trough loading receives the highest rank because it affects capacity, flow stability, power demand, and material carry. Screw rotational speed ranks second because speed controls wear, dust generation, noise, and maintenance frequency. Conveyor throughput capacity ranks third because the conveyor has a fixed 150 TPH requirement. Screw diameter ranks fourth because it controls the relationship between required capacity and operating speed. Installed motor power, hanger bearing count, and sealing performance also remain important because they affect reliability, safety, and service cost.

Table 2.4. Ranked Engineering Characteristics

Rank	Engineering Characteristic	Raw Score	Relative Weight
1	Trough loading	123	15.22
2	Screw rotational speed	90	11.14
3	Conveyor throughput capacity	81	10.02
4	Screw diameter	76	9.41
5	Installed motor power	72	8.91
6	Number of intermediate hanger bearings	69	8.54
7	Enclosure and sealing system performance	63	7.80
8	Maximum lump size	60	7.43
9	Allowable shaft torque	57	7.05
10	Shaft diameter	40	4.95
11	Manufacturing cost	39	4.83
12	Shaft deflection	38	4.70

2.1.6. Room 6: Customer Assessment of Competing Products

Room 6 compares competing conveyor solutions using the customer requirements. Martin Sprocket, Screw Conveyor Corporation, and aero-mechanical conveyors were compared because they represent common bulk material handling options. Screw Conveyor Corporation scores high in capacity, lump handling, dust-tight operation, wear resistance, and reliability. Martin Sprocket scores well in reliability and assembly simplicity. Aero-mechanical conveyors score well in low cost and low noise, but they are less aligned with the selected horizontal screw conveyor layout for abrasive limestone. This assessment confirms that the selected design needs the durability and containment advantages of an industrial screw conveyor while improving maintenance access and reducing internal obstruction.

Table 2.5: Customer Assessment of Competing Products

Customer Requirement	Martin Sprocket	Screw Conveyor Corp.	Aero-mechanical Conveyors
High conveying capacity	4	5	4
Continuous and stable material flow	3	4	3
Handle lump size and uniform feed	4	5	4
Dust-tight safe operation	4	5	3
High abrasion and wear resistance	4	5	4
Reliable operation and long service life	5	5	4
Low maintenance frequency and downtime	3	4	3
Energy efficient operation	3	4	4
Low manufacturing cost	3	3	5

Customer Requirement	Martin Sprocket	Screw Conveyor Corp.	Aero-mechanical Conveyors
Low assembly complexity and part count	5	1	3
Low noise level	1	3	4
Minimal obstruction to material flow	3	4	4

2.1.7. Room 7: Engineering Assessment of Competing Products

Room 7 compares the same competing products using engineering characteristics. Screw Conveyor Corporation scores high in screw diameter range, rotational speed suitability, throughput capacity, and lump handling. Martin Sprocket scores high in enclosure and sealing performance. Aero-mechanical conveyors score high in sealing, torque rating, shaft deflection, motor power, and noise-related performance, but that concept is not the best match for abrasive limestone handling at the required layout. The engineering assessment shows that the final design prioritizes trough loading, low screw speed, dust containment, and shaft support instead of only minimizing cost.

Table 2.6. Engineering Assessment of Competing Products

Engineering Characteristic	Martin Sprocket	Screw Conveyor Corp.	Aero-mechanical Conveyors
Screw diameter	4	5	4
Screw rotational speed	3	4	3
Conveyor throughput capacity	5	5	4
Maximum lump size	4	4	4
Trough loading	4	4	3
Enclosure and sealing system performance	5	4	5
Number of intermediate hanger bearings	3	2	3
Shaft diameter	4	4	4
Allowable shaft torque	4	3	5
Shaft deflection	3	3	4
Installed motor power	3	3	5
Manufacturing cost	3	4	2

2.1.8. Room 8: Target Values

Room 8 defines the final engineering target values for the selected screw conveyor. The original concept-level targets were refined after parametric design. Candidate B was selected as the final configuration because the 30 in screw meets the 150 TPH requirement at approximately 15.9 RPM, which reduces wear and supports the low-maintenance direction of Concept 3. The selected design uses 30B trough loading, handles 2 in limestone lumps, uses a covered dust-tight trough, and minimizes hanger bearings to reduce internal obstruction. Standard catalog drive components and fabricated steel parts control manufacturing cost and simplify replacement [1].

Table 2.7. Target Values

Engineering Characteristic	Target Values
Screw diameter	30 in
Screw rotational speed	Approximately 15.9 RPM
Conveyor throughput capacity	150 TPH
Maximum lump size	2 in
Trough loading	30B loading

Engineering Characteristic	Target Values
Enclosure and sealing system performance	Covered dust-tight trough with seals
Number of intermediate hanger bearings	0 to 1
Shaft diameter	Sized for torque and deflection
Allowable shaft torque	Greater than calculated operating torque
Shaft deflection	Controlled by shaft sizing and support layout
Installed motor power	Selected with service factor
Manufacturing cost	Reduced using standard parts and fabricated carbon steel components

The HOQ results guided the final concept and parametric design decisions. The top-ranked engineering characteristics led to the selection of a larger, slower screw instead of the smallest feasible diameter. The final 30 in screw at approximately 15.9 RPM satisfies the 150 TPH requirement, lowers wear compared with the 24 in option, avoids the higher cost of the 36 in option, and supports the selected low-maintenance concept. The final design also uses dust-tight covers, reduced hanger obstruction, wear-resistant flights, and standard drive components to meet the customer requirements identified in the HOQ.

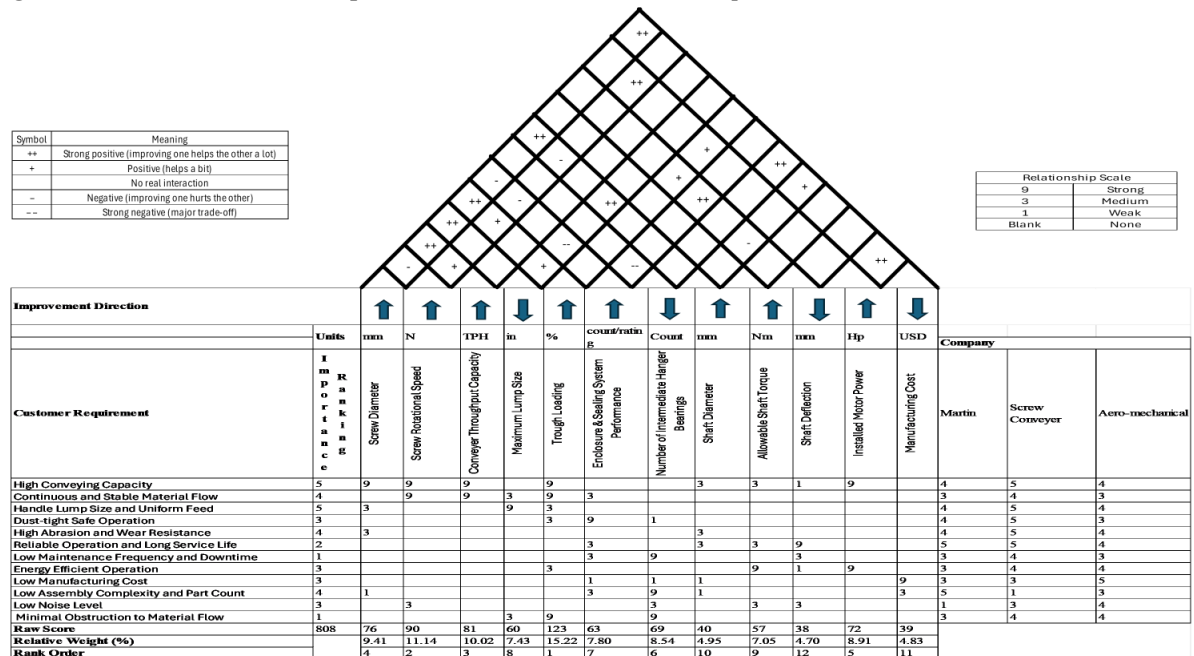


Figure 3. House of Quality.

2.2 Product Design Specification

The product design specification (PDS) defines the functional, operational, and manufacturing requirements of the screw conveyor system. These specifications serve as the foundation for concept development and evaluation. Table 2.8 provides detailed PDS.

The system is intended for industrial material handling applications, specifically for conveying crushed limestone. The design must support continuous operation, ensure dust containment, and maintain structural integrity under abrasive conditions. Additionally, the system must be compatible with standard industrial drive components such as motors and V-belt systems. Key performance requirements include a

conveying capacity of 150 TPH, handling of material with a maximum lump size of 2 inches and maintaining uniform material flow with minimal shaft deflection. From a manufacturing perspective, the design emphasizes the use of standard components, reduced machining complexity, and modular construction to facilitate maintenance and part replacement. Environmental considerations include operation in an indoor industrial setting with exposure to dust and abrasive materials, requiring appropriate sealing and wear-resistant components.

Table 2.8. Product Design Specification for Screw Conveyor

Product Identification	Market Identification
– Industrial screw conveyor – 150 TPH limestone – Continuous horizontal 30 ft distance – Industrial motor and V-belt drive compatible – Integration with upstream belt conveyor feed	– Target users: Material handling facilities – Initial application: Limestone handling system – Market need: Reliable, dust-controlled conveying – Competing products: Standard screw conveyors
Special Features	Financial Requirements
– Continuous stable flow – Dust-tight sealed trough – Abrasion-resistant components – Reduced hanger obstruction – Modular design for easy maintenance	– Reduce manufacturing cost – Use standard catalog components – Minimize machining complexity – Reduce part count
Key Performance Targets	Life Cycle Targets
– Capacity: 150 TPH – Max lump size: 2 in – Continuous operation – Low power consumption – Minimal shaft deflection – Uniform material flow	– Service life: $\geq$ Industry standard – Easy replacement of wear components – Reliable performance under continuous operation – Reduced maintenance frequency
User Training Requirements	Service Environment
– No specialized training required – Standard industrial maintenance procedures – Accessible inspection points for operators	– Material: Crushed limestone – Free flowing but abrasive – Dust generation expected – Indoor installation – Industrial ambient temperature

2.3 Physical Decomposition

The physical decomposition defines the screw conveyor system in terms of its major subsystems and corresponding components. As shown in Figure 4, the overall system is divided into six primary units which include the drive unit, conveying unit, containment unit, support unit, interface unit, and protection unit. Each unit represents a functional grouping of components that collectively perform the required conveying operation.

The drive unit consists of the motor, V-belt, reducer, and coupling, which provide the required power transmission. The conveying unit includes the screw shaft, flights, and end shaft responsible for material movement. The containment unit is composed of the trough, dust-tight cover, and access openings to ensure safe and controlled operation. The support unit includes end bearings, hanger bearings, and the

support frame to maintain structural stability. The interface unit manages material entry and exit through the inlet, feed transition, and discharge chute. The protection unit includes seals, wear liners, and hard facing elements to reduce material leakage and wear under abrasive conditions. This decomposition provides a clear mapping of the system architecture and supports further design analysis, component selection, and assembly planning.

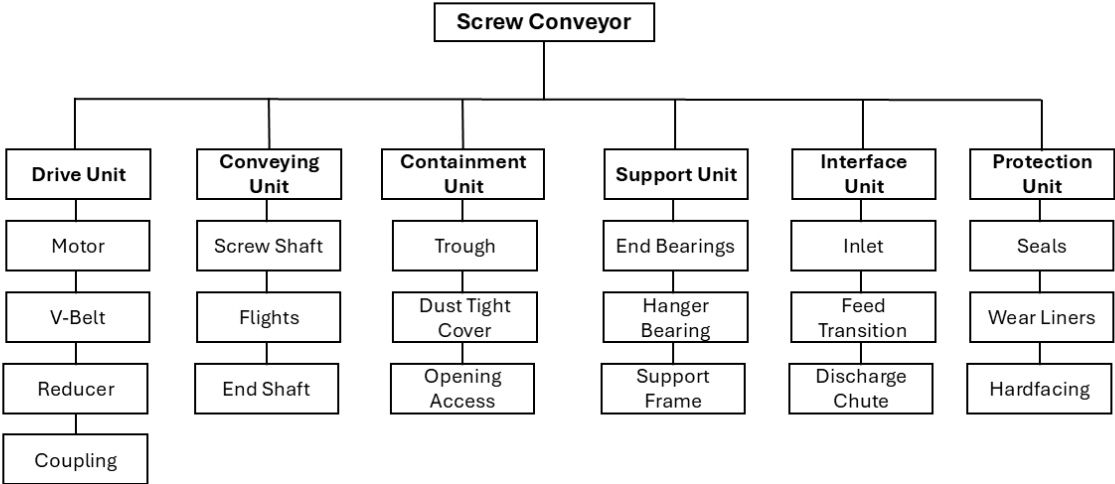


Figure 4. Physical Decomposition of the Screw Conveyor System Showing Major Units and Components.

2.4 Functional Analysis

A functional decomposition of the screw conveyor system was performed to understand the transformation of inputs into desired outputs. The system primarily converts electrical energy into mechanical energy to facilitate the transportation of material. The main functions include receiving material, containing and metering the flow, transporting material through the screw mechanism, and discharging it at the outlet. Supporting functions include dust containment, speed regulation, and torque transmission. This functional breakdown, shown in Figure 5, identifies key subsystems and ensures that all necessary operations are considered during concept development.

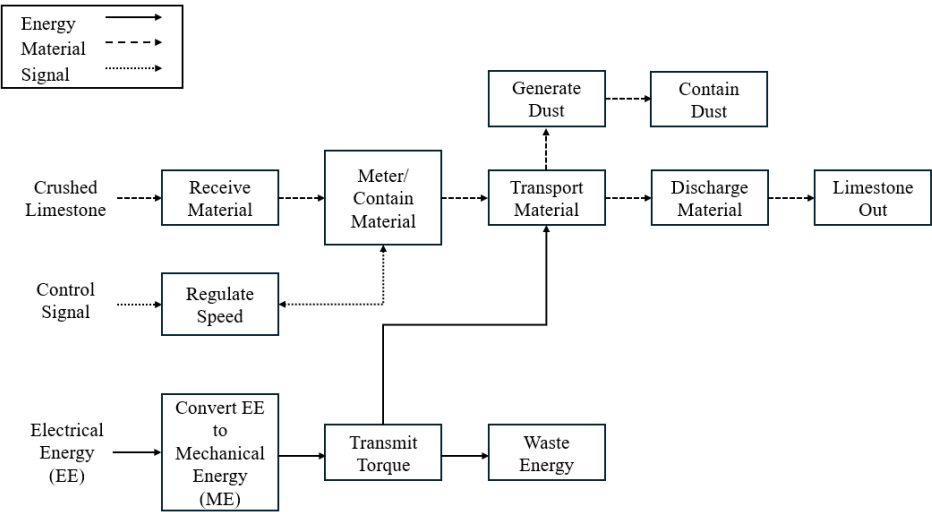


Figure 5. Functional Structure of Screw Conveyor.

2.5 Product Architecture

The screw conveyor uses a modular product architecture (Figure 6) to improve reliability, maintenance, and ease of assembly. The system is divided into four main modules: Housing & Containment, Conveying & Support, Drive & Power Transmission, and Protection & Maintenance.

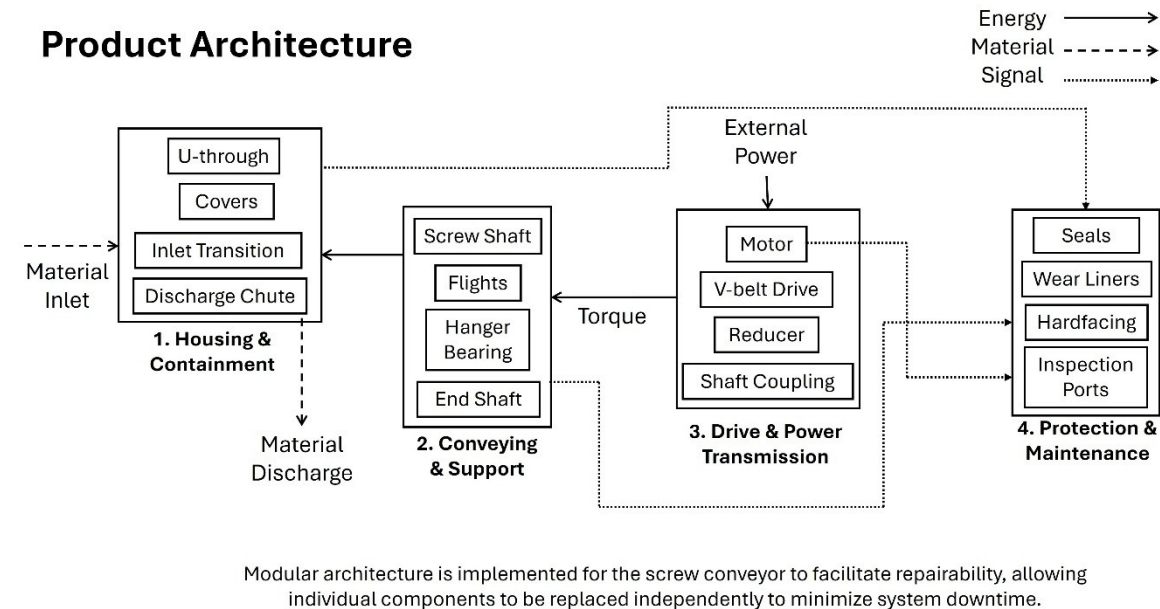


Figure 6. Product Architecture for Screw Conveyor.

2.5.1 Housing and Containment

The housing and containment subsystem includes the trough, covers, inlet transition, discharge chute, and seals. Its purpose is to contain the material, guide the flow, and reduce dust release. For limestone conveying, this subsystem is very important because dust leakage can create safety, cleanliness, and maintenance problems.

2.5.2 Conveying and Support

The conveying and support subsystem includes the flights, screw shaft, hanger bearing, and end shaft. This subsystem directly moves the limestone through the trough. It must resist torque, bending, abrasive wear, and misalignment.

2.5.3 Drive and Power Transmission

The drive subsystem includes the motor, shaft coupling, V-belt drive, and reducer. These components convert electrical energy into controlled mechanical rotation. Since the drive system is a major source of downtime if overloaded or misaligned, the design uses standard industrial components for improved reliability and easier replacement.

2.5.4 Protection and Maintenance

The protection and maintenance subsystem includes hardfacing, wear liners, inspection ports, seals, and other replaceable protective features. These components are included to extend service life and simplify maintenance. The modular arrangement allows individual parts to be inspected, repaired, or replaced without disassembling the entire system. It also supports low maintenance, reduced downtime, better wear

resistance, and future upgrades, which are important for handling abrasive crushed limestone in an industrial environment.

2.6 Morphological Chart

A morphological chart was developed to systematically generate and organize multiple design concepts for the screw conveyor system. The chart breaks down the overall function into key sub-functions and identifies alternative solutions for each function. These sub-functions include material feeding, conveying mechanism, containment, support, power transmission, and protection.

For each sub-function, multiple feasible design options were listed based on engineering judgment, industry practices, and the project requirements. This approach ensures that the concept generation process remains structured and solution-neutral while exploring a wide range of possible configurations. The combinations of these alternatives form different design concepts, which are later evaluated using a weighted decision matrix. The morphological chart therefore serves as a critical tool in linking functional analysis to concept generation and supports the selection of the most suitable design for the given application.

Table 2.9. Morphological Chart Showing Design Alternatives for Key Screw Conveyor Functions

Functional Subproblem	Option 1	Option 2	Option 3	Option 4
Handle material characteristics	Free-flowing granular	Fine powder. dusty	Abrasive	Corrosive, heat-corrosive
Select construction material	Carbon Steel	SS304	SS316	AR400, lined steel
Convey material	Standard flight	Ribbon flight	Paddle screw	Cut, folded flight
Control material flow	Standard pitch	Short / half pitch	Variable pitch	Long pitch
Contain and guide material	U-trough	Tubular	Flared trough	Rectangular trough
Introduce material into system	Open inlet	Hopper inlet	Controlled feed	Flood-fed inlet
Discharge material	End discharge	Intermediate discharge	Bottom discharge	Multiple discharge
Provide power and motion	V-belt drive	Direct drive	Gearmotor	Chain drive
Support rotating shaft	Style 226	Style 216	Style 220	Air-purged
Select bearing mechanism	Bronze	Hard iron	Ball bearing	UHMW, nylon
Enclose system (cover type)	Open cover	Bolted cover	Dust-tight cover	Hinged cover
Seal system interfaces	Plate seal	Packing gland	Waste pack	Air-purged seal
Enhance wear resistance	None	Hardfaced flights	AR steel	Replaceable liners
Enhance wear resistance	None	Hardfaced flights	AR steel	Replaceable liners

2.7 Concept Design

Based on the functional analysis and morphological chart, four design concepts were developed by combining different solutions for material handling, containment, support, and maintenance. These

concepts are summarized in table 2.10. Each concept satisfies the basic requirement of conveying crushed limestone at 150 TPH but differs in cost, dust control, wear resistance, and maintenance strategy.

2.7.1 Concept 1

It uses a conventional open U-trough configuration with standard flights, bronze bearings, and a V-belt drive. The design prioritizes low manufacturing cost and simple fabrication using standard components. However, the open configuration provides poor dust containment, and the use of standard materials results in limited wear resistance. The presence of multiple hanger bearings increases maintenance frequency and introduces potential flow obstruction. Figure 7 illustrates the physical configurations of this concept.

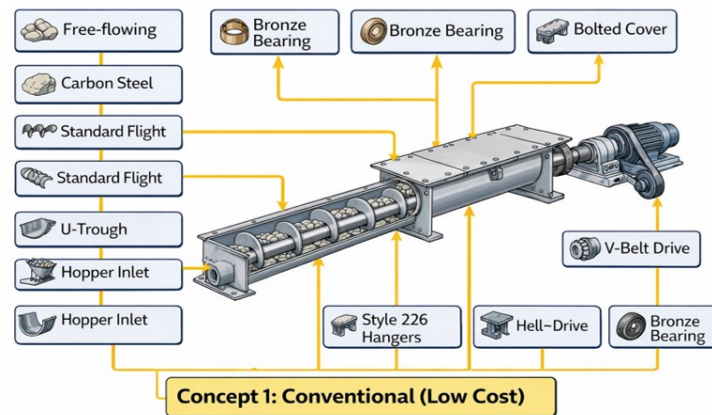


Figure 7. Concept 1 Conventional Screw Conveyor Design.

2.7.2 Concept 2

It focuses on dust containment and controlled operation. The design uses a more enclosed configuration with dust-tight covers, controlled feed inlet, and improved sealing. Stainless steel components and ball bearings improve corrosion resistance and operational smoothness. This concept reduces material loss and improves safety but introduces higher manufacturing cost, increased power consumption, and more complex assembly due to additional sealing components. Figure 8 illustrates the physical configurations of this concept.

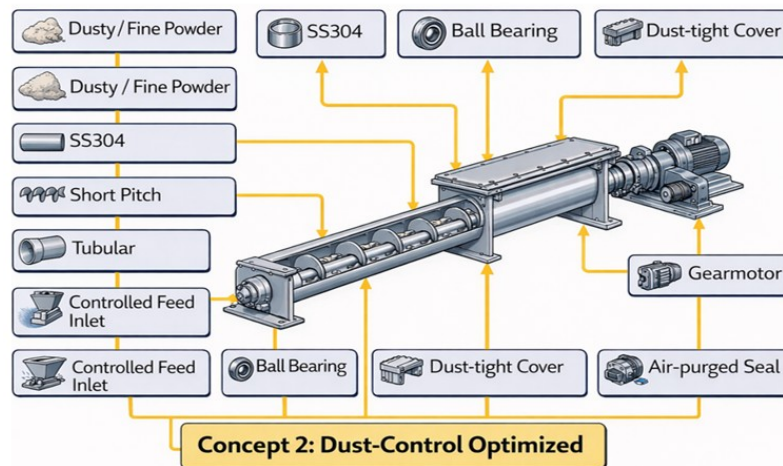


Figure 8. Concept 2 Dust Control Optimized Screw Conveyor Design.

2.7.3 Concept 3

It is developed to minimize maintenance and improve reliability under abrasive conditions. The design reduces the number of hanger bearings to decrease internal obstruction and wear points. Abrasion-resistant materials such as AR400 and hardfaced flights are used to extend service life. This concept reduces downtime and maintenance frequency but requires a stronger shaft to handle increased span and load, leading to higher initial cost and structural design complexity. Figure 9 illustrates the physical configurations of this concept.

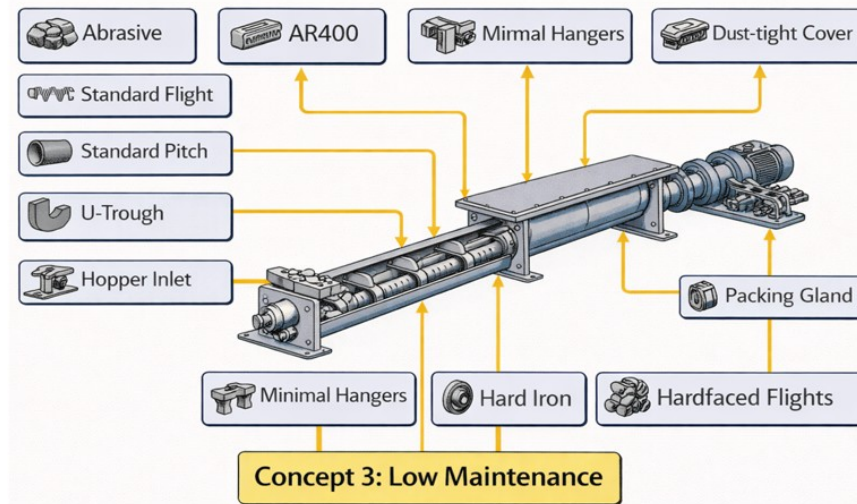


Figure 9. Concept 3 Low Maintenance Screw Conveyor Design.

2.7.4 Concept 4

It emphasizes maintainability and durability through modular construction. The design includes split-tube sections, quick-release covers, drop-bottom trough, and easy-access hanger bearings. Stainless steel construction and wear-resistant components improve service life and reduce component failure. However, this concept results in the highest manufacturing cost, increased system weight, and more complex fabrication and assembly processes. Figure 10 illustrates the physical configurations of this concept.

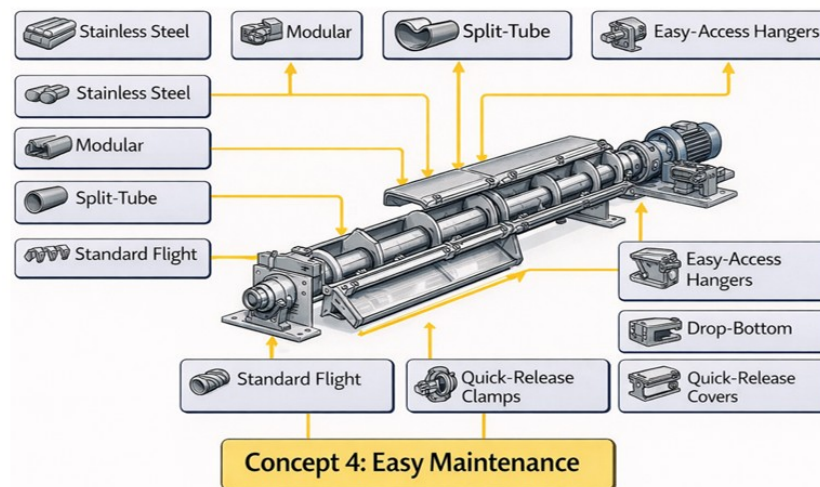


Figure 10. Concept 4 Easy Maintenance Screw Conveyor Design.

Table 2.10. Concept Comparison Summary

Functional Subproblem	Concept 1 Conventional	Concept 2 Dust-Control Optimized	Concept 3 Low Maintenance	Concept 4 Easy Maintenance
Handle material characteristics	Free-flowing	Dusty	Abrasive	Abrasive
Select construction material	Carbon Steel	SS304	AR400	Stainless Steel
Convey material	Standard flight	Standard flight	Standard flight	Standard flight
Control material flow	Standard pitch	Short pitch	Standard pitch	Standard pitch
Contain and guide material	U-trough	Tubular	U-trough	Split-tube
Introduce material into system	Hopper inlet	Controlled feed	Hopper inlet	Hopper inlet
Discharge material	End discharge	End discharge	End discharge	Intermediate discharge
Provide power and motion	V-belt drive	Gearmotor	V-belt drive	Gearmotor
Support rotating shaft	Multiple hangers (226)	Dust-tight hangers (226)	Minimal hangers	Easy-access hangers
Select bearing mechanism	Bronze	Ball bearing	Hard iron	Ball bearing
Enclose system (cover type)	Bolted cover	Dust-tight cover	Dust-tight cover	Quick-release cover
Seal system interfaces	Packing gland	Air-purged seal	Packing gland	Air-purged seal
Enhance wear resistance	None	AR coating	Hardfaced flights	Replaceable liners
Enhance wear resistance	Standard covers	Inspection ports	Modular sections	Modular, quick-release

2.8 Objective Tree

The objective tree organizes the main design goals of the screw conveyor into a clear hierarchy. The overall objective is to develop an effective screw conveyor, which is divided into three major categories: cost, service, and performance. Cost includes manufacturing cost, maintenance cost, and material cost. Service includes repairability, easy assembly, and low downtime. Performance includes durability, loading capacity, and dust tightness (Figure 11).

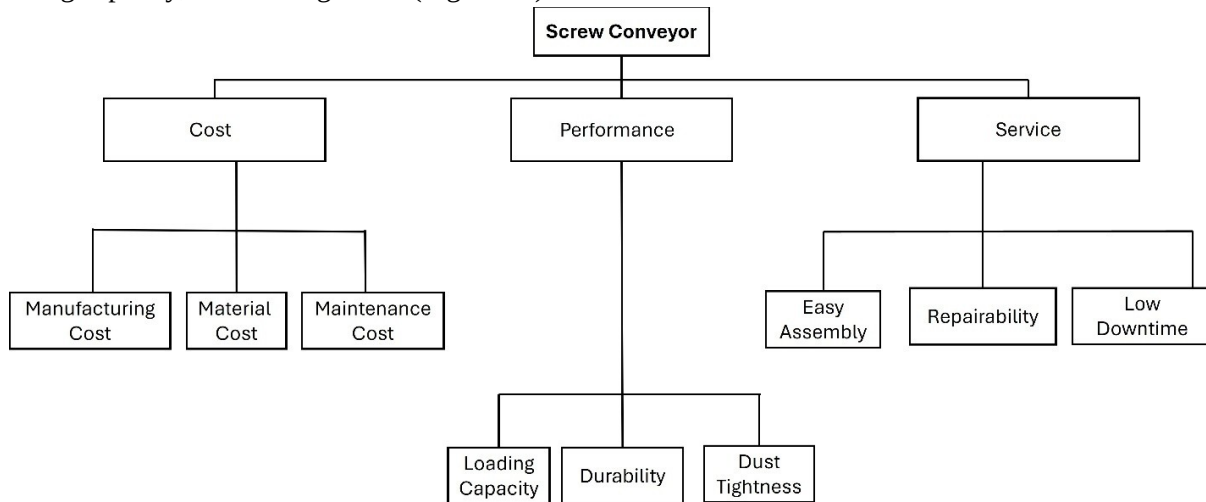


Figure 11. Objective Tree.

The objective tree breaks down the main design goal of the screw conveyor into three weighted categories: Cost ($O_{11} = 0.3$), Service ($O_{12} = 0.4$), and Performance ($O_{13} = 0.3$). Service has the highest weight, showing that reparability, easy assembly, and low downtime are the main priorities for this design.

Each main objective is further divided into sub-objectives. Cost includes manufacturing, maintenance, and material cost; service includes reparability, assembly, and downtime; and performance includes durability, loading capacity, and dust tightness. This tree helps compare design alternatives by showing which factors are most important in selecting the best screw conveyor concept.

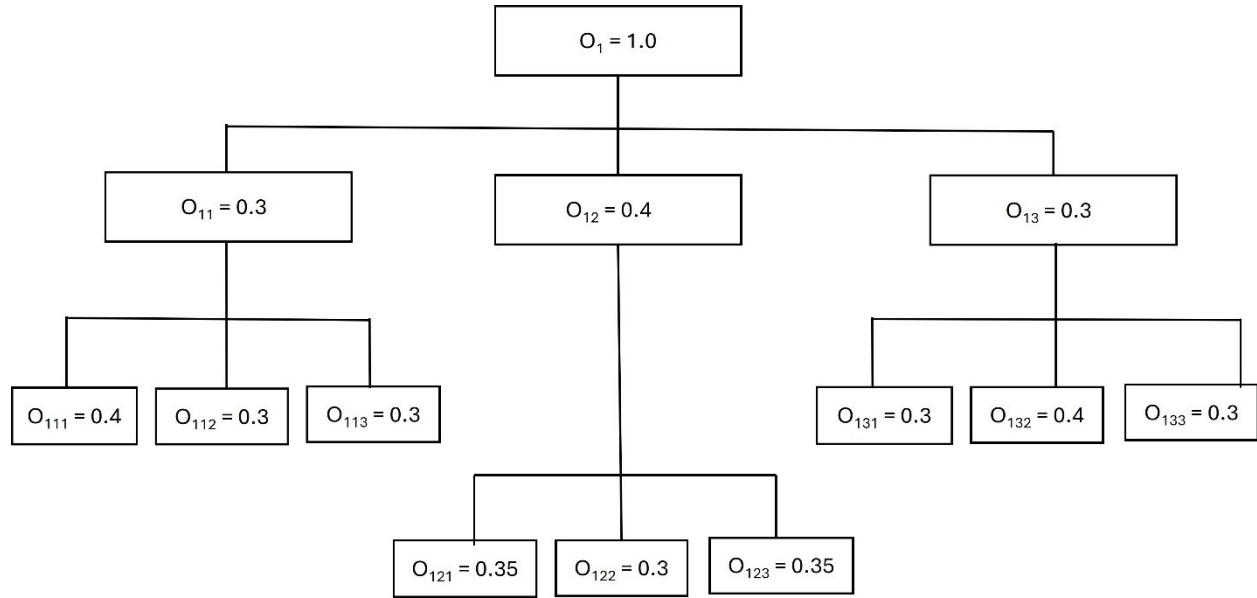


Figure 12. Objective Tree with Weighted Decision.

This structure is useful because it shows the design is not evaluated only by technical performance. It must also be practical to manufacture and maintain.

2.9 Concept Generation and Selection

Multiple design concepts were developed to meet the functional and performance requirements of the screw conveyor system. These concepts varied in terms of sealing mechanisms, wear protection strategies, manufacturing complexity, and structural configuration.

To systematically evaluate the concepts, a weighted decision matrix was employed. Evaluation criteria included conveying capacity, dust containment, wear resistance, reliability, maintenance accessibility, manufacturing cost, assembly simplicity, and energy efficiency.

Each criterion was assigned a weight based on its importance, and the concepts were rated on a scale from 1 to 5. The weighted scores were then used to determine the most suitable design.

Table 2.11. Weighted Decision Matrix for Concept Selection

Selection Criteria	Weight	Units	Concept 1			Concept 2			Concept 3			Concept 4		
			Magnitude	Score	Rating	Magnitude	Score	Rating	Magnitude	Score	Rating	Magnitude	Score	Rating
Conveying capacity	0.18	ton/hr	140	3	0.54	150	5	0.90	150	5	0.90	160	5	0.90
Dust containment	0.15	qualitative	Moderate sealing	3	0.45	Excellent Sealing	5	0.75	Good Sealing	4	0.60	Good sealing	4	0.60
Wear resistance	0.14	qualitative	Standard	3	0.42	Standard + limited wear protection	3	0.42	Improved wear protection	4	0.56	Heavy-duty wear protection	5	0.70
Reliability	0.12	years	6	3	0.36	7	4	0.48	9	5	0.60	8	4	0.48
Maintenance accessibility	0.10	qualitative	Moderate	3	0.30	Moderate	3	0.30	Excellent	4	0.40	Moderate	3	0.30
Flow obstruction	0.08	qualitative	Moderate	3	0.24	good	4	0.32	good	4	0.32	Moderate	3	0.24
Manufacturing cost	0.09	\$	18,000	5	0.45	24,000	3	0.27	21,000	4	0.36	28,000	2	0.18
Assembly simplicity	0.05	Parts Count	22	4	0.20	28	3	0.15	24	4	0.20	31	2	0.10
Energy efficiency	0.04	hp	30	4	0.16	32	3	0.12	29	5	0.20	34	3	0.12
Constraint compatibility	0.05	good	4	4	0.20	good	4	0.20	Excellent	5	0.25	Fair	3	0.15
	1.00				3.32			3.91			4.39			3.77

Concept 3 achieved the highest overall score of 4.39, indicating a balanced performance across all criteria, particularly in wear resistance, reliability, and manufacturability. Therefore, Concept 3 was selected for further development.

3. Preliminary Design Review

The preliminary design review focuses on refining the selected concept through detailed engineering decisions related to materials, manufacturing processes, and assembly considerations. This section evaluates the design from a manufacturability and practicality standpoint, ensuring that the selected configuration can be produced efficiently and reliably under industrial conditions. Key analyses include parametric, material and process selection with justification, design for manufacturing and design for assembly evaluations, and tolerance stack-up analysis for critical components. These analyses ensure that the design meets performance requirements while minimizing cost, simplifying production, and supporting ease of assembly and maintenance.

3.1 Parametric Design

Parametric design is used to systematically develop the screw conveyor by defining key design variables and evaluating their influence on system performance. The process begins with formulation of the design problem, followed by identification of engineering characteristics, design variables, and operating conditions. Alternative designs are then generated and screened using capacity and feasibility constraints, and the feasible options are analyzed based on performance metrics such as speed, wear, and maintenance requirements. A comparative and weighted evaluation is performed to select the optimal configuration, after which the selected design is refined through parameter optimization. As shown in Figure 13, this process is iterative, where the refinement step can feed back into the alternative design stage to further improve performance and ensure all design objectives are satisfied.

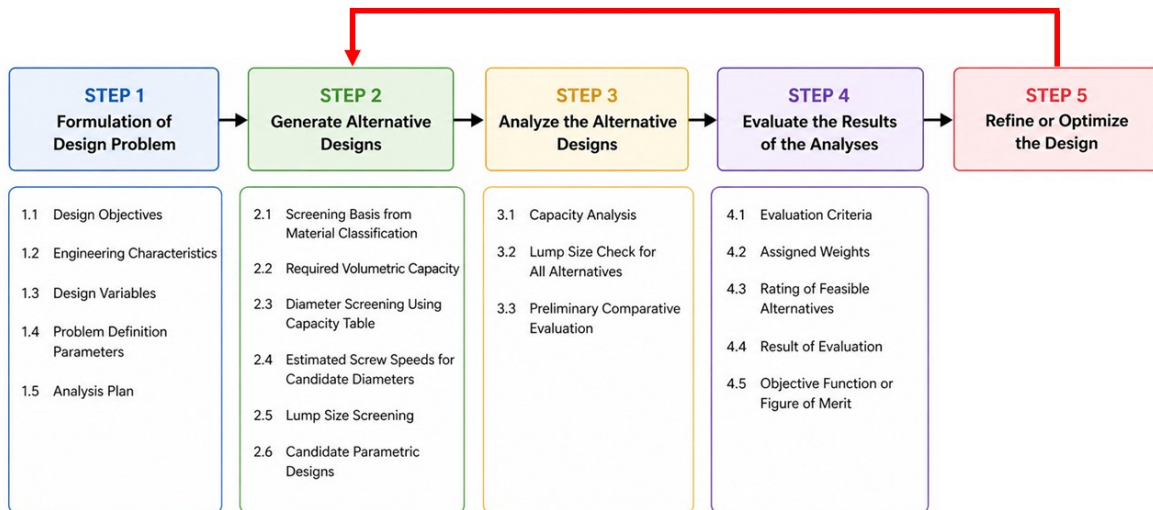


Figure 13. Parametric Design Process Showing Sequential Steps.

3.1.1 Step 1: Formulation of Design Problem

The formulation of the design problem defines the foundation of the parametric design process by establishing the objectives, performance measures, and key variables of the system. This step organizes the problem into clearly defined sub-components, including design objectives, engineering characteristics, design variables, and problem parameters, which are presented in the following subsections.

3.1.1.1. Design Objectives

The screw conveyor is designed to transport crushed limestone at a capacity of 150 TPH over a horizontal distance of 30 ft. The design is based on Concept 3, which emphasizes low maintenance and minimal internal obstruction. The system must ensure continuous and stable material flow while maintaining reliability under abrasive operating conditions. The design must also satisfy constraints related to dust containment, manufacturability, and structural integrity.

3.1.1.2. Engineering Characteristics

The performance of the screw conveyor design is evaluated using engineering characteristics summarized in Table 3.1, derived from the Product Design Specification and the House of Quality. These characteristics define measurable parameters that are used to assess system performance, structural behavior, and operational efficiency.

Table 3.1. Engineering Characteristics (Performance Metrics)

Parameter	Unit	Description
Conveying capacity	TPH	Material throughput of the conveyor
Screw rotational speed	RPM	Controls material flow rate and affects wear
Screw diameter	in or mm	Determines conveying capacity
Shaft torque	Nm	Load transmitted through the shaft
Shaft deflection	mm	Indicator of structural performance
Power requirement	HP	Energy required to drive the system
Wear resistance	qualitative	Resistance to abrasion due to limestone
Dust containment effectiveness	qualitative	Ability to prevent material leakage
Manufacturing cost	USD	Cost associated with production
Assembly complexity and maintenance	qualitative	Ease of assembly, inspection, and maintenance

3.1.1.3. Design Variables

The design variables (Table 3.2) define the key parameters controlled and optimized to achieve the desired performance of the screw conveyor. These variables influence conveying capacity, structural integrity, wear behavior, and maintenance requirements. The selected variables are based on standard screw conveyor design practices and are chosen to align with the low-maintenance objective of the selected concept.

Table 3.2. Design Variables

Variable	Symbol	Unit	Description
Screw diameter	(D)	mm or in	Determines conveying capacity
Screw speed	(N)	RPM	Controls material flow rate
Screw pitch	(P)	mm or in	Affects flow behavior
Shaft diameter	(d _s)	mm	Determines strength and deflection
Trough loading	(η)	%	Fill level of conveyor
Number of hangers	(n _h)	–	Affects obstruction and maintenance
Motor power	HP	HP	Required to drive the system
Flight thickness	(t _f)	mm	Influences wear resistance
Trough thickness	(t _t)	mm	Provides structural strength

3.1.1.4. Problem Definition Parameters

The problem definition parameters, as summarized in Table 3.3, specify the operating conditions and constraints under which the screw conveyor is designed. These parameters define the material characteristics, system requirements, and environmental conditions that directly influence design decisions and performance evaluation.

Table 3.3. Problem Definition Parameters

Parameter	Value	Description
Material	Crushed limestone	Abrasive, free-flowing material
Capacity requirement	150 TPH	Required conveying rate
Conveyor length	30 ft	Horizontal transport distance
Orientation	Horizontal	No inclination considered
Maximum lump size	2 in	Influences screw diameter selection
Feed condition	Uniform feed from belt conveyor	Ensures steady material input
Environment	Dusty indoor industrial operation	Requires sealing and wear protection
Design requirement	Minimize hanger bearings and maintenance	Supports low-maintenance concept

3.1.1.5. Analysis Plan

The analysis plan, as summarized in Table 3.4, defines the sequence of calculations and evaluations used to develop and validate the screw conveyor design. The design is based on standard screw conveyor design methodology and incorporates both analytical calculations and engineering judgment to ensure performance, reliability, and manufacturability. The analysis follows a structured approach to ensure that all critical aspects of the design are addressed and that the final configuration meets the required performance and operational constraints.

Table 3.4. Analysis Plan

Analysis Step	Description
Capacity and speed calculations	Determination of conveying capacity and screw rotational speed based on CEMA standards
Screw diameter selection	Selection based on required throughput and maximum lump size
Power and horsepower calculations	Estimation of power required to drive the system
Shaft torque evaluation	Calculation of torque transmitted through the shaft
Shaft deflection analysis	Evaluation of structural behavior under load
Component selection	Selection of bearings, shaft, flights, and trough components
DFM and DFA considerations	Evaluation of manufacturability and ease of assembly

3.1.2 Step 2: Generate Alternative Designs

This step focuses on developing feasible design alternatives based on the defined problem parameters and engineering characteristics. Standard screw conveyor design data and material classification are used to establish screening criteria, which guide the selection of suitable screw diameters, speeds, and configurations. Multiple candidate designs are generated by varying key parameters while ensuring that all options satisfy the required capacity, material handling characteristics, and operational constraints.

3.1.2.1. Screening Basis from Material Classification

Material classification is used to establish the initial screening criteria for selecting suitable screw conveyor parameters. Based on standard screw conveyor design tables, crushed limestone is classified as an abrasive and free-flowing material, and its properties are used to determine appropriate design factors such as capacity, speed, and component selection. These are summarized in Table 3.5. These parameters provide the basis for screening feasible screw diameters and operating speeds for the given application.

Table 3.5. Analysis Plan

Parameter	Value	Description
Material	Limestone, crushed	Free-flowing, abrasive material
Bulk density	85–90 lb/ft ³	Used for capacity calculations
Material code	DX-36	Standard classification from design tables
Bearing selection	H	Recommended bearing type for material
Component series	2	Standard conveyor component classification
Material factor ()	2.0	Used in power and capacity calculations
Recommended trough loading	30B	Determines allowable fill level

3.1.2.2. Required Volumetric Capacity

The required conveying rate for the screw conveyor is 150 tons per hour. Converting to mass flow rate: Using the bulk density range of crushed limestone, 85–90 lb/ft³, the corresponding volumetric flow rate is calculated as:

Therefore, the required volumetric capacity lies in the range:

For preliminary design and screening purposes, a representative mid-range value is selected:

This value is used in subsequent calculations for diameter selection and speed estimation.

3.1.2.3. Diameter Screening Using Capacity Table

Based on the recommended trough loading condition of 30B, standard horizontal screw conveyor capacity tables are used to compare available capacities with the required volumetric flow rate of 3430 ft³/hr, as summarized in Table 3.6. This comparison is used to identify feasible screw diameters that can meet the design requirement.

Table 3.6. Diameter Screening Using Capacity Table

Screw Dia (in)	Max Capacity (ft ³ /hr)	Required Capacity (3430 ft ³ /hr)	Feasibility
18	2025	< Required	Not feasible
20	2500	< Required	Not feasible
24	4360	≥ Required	Feasible
30	7560	≥ Required	Feasible
36	11064	≥ Required	Feasible (oversized)

The results show that the minimum acceptable screw diameter is 24 in. Diameters of 24 in, 30 in, and 36 in satisfy the required capacity and are considered feasible options. Among these, the 24 in diameter represents the most economical initial choice, while larger diameters provide additional capacity at the expense of increased size and cost.

3.1.2.4. Estimated Screw Speeds for Candidate Diameters

Using standard screw conveyor capacity tables for 30B trough loading, the capacity per RPM is obtained for each feasible screw diameter. These values are used to estimate the required rotational speed to achieve the target volumetric flow rate of 3430 ft³/hr. All calculated speeds are below the maximum recommended operating limits provided in standard design catalogs. Therefore, all three diameters remain feasible candidates for further evaluation.

Table 3.7. Capacity per RPM and Estimated Screw Speeds

Screw Diameter (in)	Capacity per RPM (ft ³ /hr per RPM)	Required Speed (RPM)
24	109	31.5
30	216	15.9
36	368.8	9.3

3.1.2.5. Lump Size Screening

The maximum lump size of the material is specified as 2 in. Based on standard lump size limitation tables, a screw diameter as small as 12 in can handle a 2 in lump under Class II conditions. Since the feasible screw diameters identified for this design range from 24 in to 36 in, lump size does not impose a limiting constraint on the selection. Therefore, the final selection of screw diameter is governed primarily by required throughput, maintenance considerations, and overall system cost rather than lump size limitations.

3.1.2.6. Candidate Parametric Designs

Based on the screening results for capacity, screw diameter, speed, and lump size, three candidate parametric designs are selected for further detailed analysis and are summarized in Table 3.8. These candidates represent different design trade-offs between cost, performance, and maintenance, allowing comparison across a range of feasible configurations.

Table 3.8. Candidate Parametric Designs

Candidate	Screw Dia	Estimated Speed	Preliminary Shaft Selection	Hanger Arrangement	Design Intent
A	24 in	31.5 RPM	Standard heavy shaft	1 hanger	Smallest feasible size with lower material cost
B	30 in	15.9 RPM	Larger shaft	0–1 hanger	Balanced low-speed and reduced maintenance design
C	36 in	9.3 RPM	Largest shaft	0 hanger preferred	Maximum maintenance reduction with lowest operating speed

3.1.3 Step 3: Analyze the Alternative Designs

In this step, the candidate parametric designs are evaluated using engineering analysis to verify their performance and feasibility. Key parameters such as capacity, speed, power, and structural behavior are

assessed to ensure that each design satisfies the required operating conditions. This analysis provides the basis for comparing alternatives and selecting the most suitable design.

3.1.3.1. Capacity Analysis

The capacity performance of each candidate design is evaluated using standard screw conveyor capacity data for 30B trough loading and is summarized in Table 3.9. The capacity at 1 RPM, maximum recommended capacity, and allowable speed limits are compared with the required operating conditions to verify feasibility.

Table 3.9. Capacity Analysis of Candidate Designs

Parameter	A (24 in)	B (30 in)	C (36 in)
Capacity at 1 RPM (ft ³ /hr)	109	216	368.8
Maximum recommended capacity (ft ³ /hr)	4360	7560	11064
Maximum recommended speed (RPM)	40	35	30
Required screw speed (RPM)	31.5	15.9	9.3

All three candidates satisfy the required volumetric capacity of 3430 ft³/hr while operating below the maximum recommended speed limits. This confirms that each design is feasible from a capacity and operational standpoint and can be considered for further analysis. Although all three candidates meet the required capacity and operate within allowable speed limits, their performance differs significantly in terms of operational behavior and maintenance requirements. Candidate A operates at the highest speed, which can lead to increased wear, higher maintenance needs, and greater sensitivity to flow instability. It is also more likely to require additional hanger support, resulting in higher internal obstruction. Candidate B provides a more balanced design, with moderate speed, improved wear characteristics, and reduced hanger requirements, leading to more stable operation. Candidate C operates at the lowest speed, offering the best performance in terms of wear reduction, flow stability, and minimal internal obstruction. It also aligns most closely with the design goal of reducing maintenance by minimizing or eliminating hanger bearings. However, this advantage comes with increased size, weight, and cost. Overall, Candidate C demonstrates the best alignment with the project objectives, while Candidate B remains a strong compromise between performance and cost.

3.1.3.2. Lump Size Check for All Alternatives

The maximum lump size of the material is specified as 2 in. According to standard lump size limitation tables, a screw diameter as small as 12 in can accommodate a 2 in lump under Class II conditions. Since all three candidate designs have screw diameters of 24 in or greater, lump size does not impose any restriction on the selection. Therefore, all candidates satisfy the lump size requirement and remain valid options for further evaluation.

3.1.3.3. Preliminary Comparative Evaluation

A preliminary comparison of the candidate designs is performed based on capacity, operating speed, maintenance requirements, and cost considerations, and is summarized in Table 3.10.

This evaluation provides a basis for selecting the most suitable design alternative.

Table 3.10. Preliminary Comparative Evaluation of Candidates

Option	Screw Dia	Required Speed	Capacity	Maintenance	Cost	Status
--------	-----------	----------------	----------	-------------	------	--------

A	24 in	31.5 RPM	Feasible	Less favorable	Lower	Feasible
B	30 in	15.9 RPM	Feasible	Good	Moderate	Feasible
C	36 in	9.3 RPM	Feasible	Best	Highest	Feasible

All three candidates satisfy the required capacity and remain feasible. However, their performance differs in terms of operational efficiency and design objectives. Candidate A operates at a higher speed, which may result in increased wear and higher maintenance requirements despite its lower cost. Candidate C offers the lowest operating speed and best maintenance performance, but it is oversized and results in higher material and manufacturing cost. Candidate B provides the most balanced solution, achieving the required throughput while operating at a moderate speed that reduces wear and maintenance demands. It also supports the design goal of minimizing hanger bearings and internal obstruction without significantly increasing cost. Therefore, Candidate B is selected as the final design for further development.

3.1.4 Step 4: Evaluate the Results of the Analyses

In this step, the feasible design alternatives are evaluated using a structured decision-making approach. A weighted evaluation matrix is developed to compare the designs based on key performance, maintenance, and cost criteria. This ensures that the final selection aligns with the project objectives and operational requirements.

3.1.4.1. Evaluation Criteria

A set of evaluation criteria is defined to assess the performance and suitability of the candidate designs. These criteria are selected based on the functional requirements, material characteristics, and design objectives established earlier in the study and is summarized in table 3.11.

Table 3.11. Evaluation Criteria and Justification

Criterion	Reason for Inclusion
Capacity adequacy	Must reliably deliver 150 TPH
Operating speed	Lower speed reduces wear and maintenance
Maintenance	Key objective of selected concept
Wear resistance	Limestone is abrasive
Hanger reduction	Fewer hangers reduce obstruction and downtime
Manufacturing cost	Project objective includes cost reduction
Overall concept fit	Must align with Concept 3

3.1.4.2. Assigned Weights

The evaluation criteria are assigned weights, as summarized in Table 3.12, to reflect their relative importance in the design selection process. Greater emphasis is placed on maintenance performance and capacity adequacy, consistent with the project objective of achieving a low-maintenance and reliable design.

The weights are normalized so that their total equals unity.

Table 3.12. Assigned Weights for Evaluation Criteria

Criterion	Weight
Capacity adequacy	0.20

Operating speed	0.15
Maintenance performance	0.20
Wear resistance tendency	0.15
Hanger reduction compatibility	0.15
Manufacturing cost	0.10
Overall concept fit	0.05
Total	1.00

3.1.4.3. Rating of Feasible Alternatives

The feasible design alternatives are evaluated using a weighted scoring approach based on the previously defined criteria and assigned weights. Each alternative is rated on a scale of 1 to 5, where 1 represents poor performance and 5 represents excellent performance. The ratings are assigned based on key design priorities, including lower operating speed, reduced maintenance requirements, minimal hanger usage, and cost effectiveness. The ratings are summarized in Table 3.13.

Table 3.13. Rating of Feasible Alternatives

Criterion	Weight	Candidate A (24 in)	Candidate B (30 in)	Candidate C (36 in)
Capacity adequacy	0.20	4	5	5
Operating speed	0.15	3	5	5
Maintenance	0.20	3	5	5
Wear resistance	0.15	3	5	5
Hanger reduction	0.15	3	5	5
Manufacturing cost	0.10	5	4	2
Overall concept fit	0.05	3	5	4

3.1.4.4. Result of the Alternatives

The total weighted scores for each candidate are calculated as follows:

- Candidate A: 3.40
- Candidate B: 4.90
- Candidate C: 4.65

Candidate B achieves the highest overall score, demonstrating the best balance between performance, maintenance reduction, and cost. While Candidate C performs strongly in maintenance-related aspects, it is penalized by higher cost and potential oversizing. Candidate A, although economical, operates at higher speeds, leading to increased wear and maintenance.

Candidate B (30 in screw diameter) is selected as it provides the most balanced and optimal solution aligned with the design objectives.

3.1.4.5. Objective Function

The weighted score is used as the figure of merit to compare the feasible design alternatives. This approach consolidates multiple performance criteria, including capacity adequacy, operating speed, maintenance requirements, wear resistance, manufacturability, and cost, into a single quantitative metric. The objective of the evaluation is to maximize the weighted score, such that the best design is identified as the one with the highest overall value.

Based on the weighted evaluation, Candidate B achieves the highest score of 4.90, indicating superior overall performance and alignment with the design objectives. Therefore, Candidate B is selected as the final design, as it provides the most balanced solution in terms of efficiency, reliability, maintenance, and cost.

3.1.5 Step 5: Refine or Optimize the Design

Following the evaluation stage, all candidate designs were found to be feasible, with Candidate B (30 in screw diameter) identified as the best overall option. This step focuses on refining the selected design by optimizing key parameters to enhance performance while satisfying all operational and design constraints. The refinement process targets reduced wear, improved flow stability, minimized maintenance requirements, and better manufacturability, ensuring a balanced and practical final configuration. This is summarized in Table 3.14.

Table 3.14. Final Design Optimization Decisions

Design Aspect	Optimization Decision	Purpose
Selected Design	Candidate B (30 in screw)	Best overall performance
Screw Speed	~16 RPM	Meets capacity, reduces wear
Hanger Bearings	0–1	Minimize obstruction & maintenance
Shaft Design	Sized for torque & deflection	Ensure structural integrity
Wear Protection	Hardfacing / liners	Improve durability
Components	Standard catalog parts	Improve manufacturability
Final Outcome	Optimized configuration	Balance performance, cost, maintenance

3.2 Material Selection

Material selection was completed after the operating conditions and main conveyor geometry were defined. The selected conveyor handles crushed limestone at 150 TPH over a 30 ft horizontal distance. The material is abrasive, dusty, and free-flowing, so the selected materials must support wear resistance, strength, manufacturability, and reasonable cost. The main components considered in this selection are the screw flights, shaft, trough, cover, and hanger bearing.

3.2.1. Ashby Chart Screening

Ashby chart is used as the first screening step to compare major material families. The chart helped narrow the selection toward metallic materials because the conveyor requires high stiffness, strength, wear resistance, and manufacturability. Polymer and composite materials were not selected because the conveyor is a large industrial machine handling abrasive crushed limestone. Stainless steel was considered during screening, but corrosion resistance is not the main design driver for this application, and stainless steel increases material cost. Steel-based materials were therefore selected for detailed comparison.

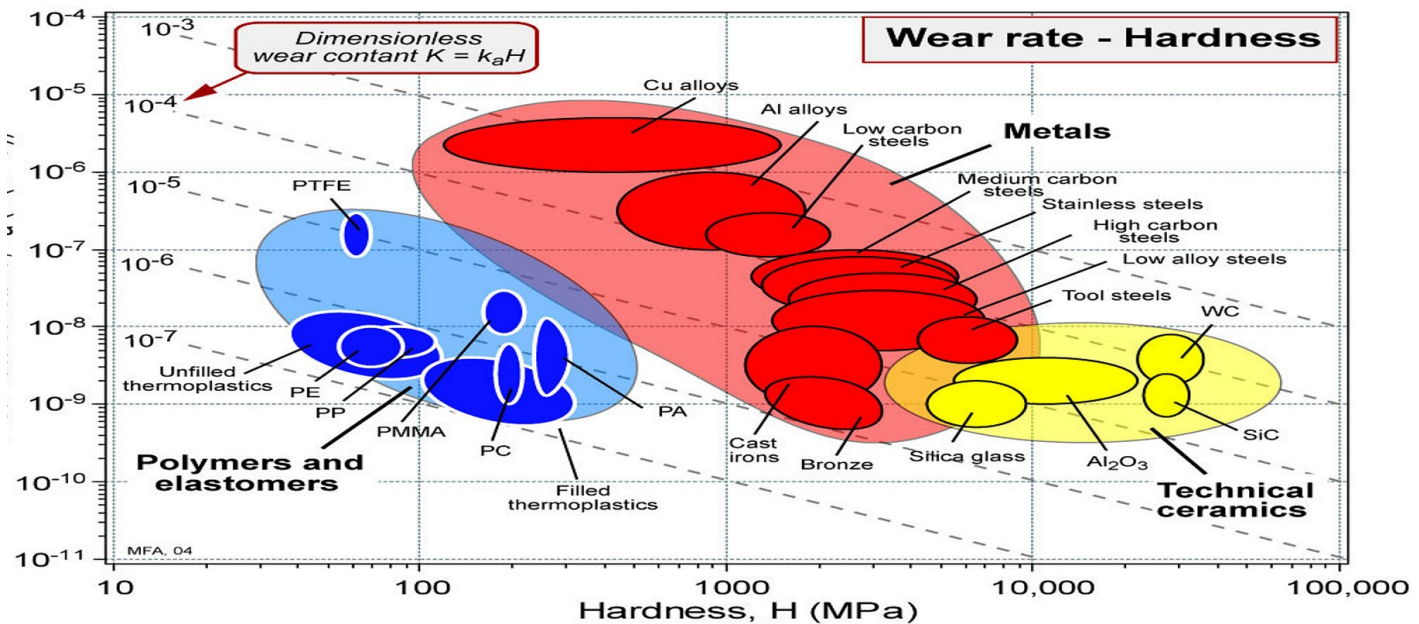


Figure 14. Ashby Chart Used for Preliminary Material Screening of Screw Conveyor Components.

The Ashby chart has been used only for early screening. Final selection was completed using component function, limestone abrasion, material properties, cost, and manufacturing suitability. The screw flights require the highest wear resistance because they directly move the limestone. The shaft requires strength and machinability because it transfers torque. The trough and cover require economical forming and welding. The hanger bearing requires wear resistance because it operates near dust and abrasive particles.

3.2.2. Material Screening for Major Conveyor Components

Table 3.15. Material Screening for Major Components

Component	Main Criteria	Candidate Materials
Screw flights	Wear resistance, strength, cost	AR225 steel, AR235 steel, AR400 steel, 304 stainless steel
Screw shaft	Strength, stiffness, machinability, cost	Carbon steel, 1045 steel, AR400 steel, 304 stainless steel
Trough and cover	Manufacturability, cost, adequate strength	Carbon steel, 304 stainless steel
Hanger bearing	Wear resistance, durability, maintenance	Hard iron, bronze, nylon

The material screening shows that one material is not suitable for every part. The screw flights need a harder wear-resistant material, while the trough and cover benefit more from low-cost steel fabrication. The shaft requires a material that is strong but still easy to machine. The hanger bearing requires a material that can survive dusty abrasive service.

3.2.3. Material Property Comparison

Table 3.16. Material Property Comparison

Material	Wear Resistance	Strength	Manufacturability	Cost	General Suitability
AR225 steel	Moderate	Moderate	Good	Low to moderate	Suitable for moderate wear components
AR235 steel	Moderate	Moderate	Good	Moderate	Suitable for moderate abrasion service
AR400 steel	High	High	Moderate	Higher	Best for direct abrasive contact
1045 steel	Moderate	High	Good	Moderate	Best for shafts and machined torque parts
Carbon steel	Low to moderate	Moderate	High	Low	Best for large fabricated parts
304 stainless steel	Moderate	Moderate	Good	High	Not preferred because corrosion is not dominant
Hard iron	High	Good	Moderate	Moderate	Suitable for hanger bearing wear service
Bronze	Moderate	Moderate	Good	High	Less suitable for abrasive limestone dust
Nylon	Low	Low to moderate	Good	Low	Not suitable for abrasive dusty service

This comparison supports the use of steel-based materials for the major conveyor structure. AR400 provides the best wear resistance for the screw flights. 1045 steel provides the best balance for the shaft because it combines strength and machinability. Carbon steel is the best choice for the trough and cover because these parts are large fabricated components. Hard iron is the most suitable choice for the hanger bearing because it provides better wear resistance in dusty abrasive service than bronze or nylon.

3.2.4. Weighted Decision Matrix for Screw Flights

The screw flights are the most important wear component in the conveyor. A weighted decision matrix compared AR225 steel, AR235 steel, AR400 steel, and 304 stainless steel. Wear resistance received the highest weight because limestone abrasion controls the life of the screw flights. Yield strength also received a high weight because the flights experience continuous loading during operation.

AR400 steel received the highest score with a total of 4.20. It is selected for the screw flights because it provides the best combination of abrasion resistance and strength for crushed limestone service. Although AR400 costs more than AR225 and AR235, the wear benefit is more important for the flights because they are the main material-moving surfaces. Stainless steel is not selected because corrosion is not the dominant issue in this application, and its higher cost does not provide enough benefit for crushed limestone.

Table 3.17. Screw Flight Material Criteria Weights

Criteria	Weight
Wear resistance	0.35
Yield strength	0.20
Ultimate tensile strength	0.10
Manufacturability	0.15
Cost	0.15
Density	0.05
Total	1.00

Table 3.18. Screw Flight Material Rating

Material	Wear Resistance	Yield Strength	UTS	Manufacturability	Cost	Density	Total Score
AR225 steel	3	3	3	4	4	3	3.35
AR235 steel	3	3	3	4	4	3	3.35
AR400 steel	5	5	5	3	3	3	4.20
304 stainless steel	2	2	2	4	1	3	2.25

3.2.5. Screw Flights

The screw flights use AR400 steel. This choice directly supports the abrasive limestone requirement. The flights push limestone through the trough, so their edges and surfaces experience repeated sliding contact with limestone particles. Softer materials lose thickness faster, which reduces conveying efficiency and increases maintenance frequency. AR400 gives the selected design a longer wear life in the highest-contact area of the conveyor.

3.2.6. Screw Shaft, Drive Shaft, Coupling Shaft, and End Shaft

The screw shaft, drive shaft, coupling shaft, and end shaft use 1045 steel. These parts transmit torque from the drive system to the screw and keep the rotating assembly aligned. The shaft material needs good strength and stiffness, but it also needs practical machining behavior for keyways, shaft steps, and bearing fits. 1045 steel provides a good balance of strength, machinability, stiffness, and cost. AR400 is not selected for the shaft because its hardness makes machining more difficult and the shaft does not experience the same abrasive contact as the screw flights.

3.2.7. Trough

The trough uses carbon steel. The trough contains limestone and supports the cover, inlet, discharge, and end components. Carbon steel is suitable because the trough is a large fabricated part that requires cutting, bending, and welding. It also keeps cost lower than stainless steel or full AR400 construction. Since limestone is abrasive, local wear zones use liners or added thickness instead of making the full trough from expensive abrasion-resistant plate. This keeps the main trough economical while still allowing wear protection in the highest-contact areas.

3.2.8. Covers and Cover with Inlet

The covers and cover with inlet use carbon steel. These components enclose the trough, control dust release, and provide removable access for inspection. Carbon steel is practical because the covers are

sheet-metal parts with simple bends and bolted edges. They do not carry the same abrasive load as the screw flights, so AR400 is not needed for the full cover system. The cover design supports dust-tight operation while keeping fabrication simple and cost controlled.

3.2.9. Hanger Bearing

The hanger bearing uses hard iron. This part supports the rotating shaft at an internal or intermediate location and operates close to dust and abrasive fines. Hard iron has high surface hardness and better wear resistance than bronze or nylon for this limestone application. Bronze has good bearing behavior, but the cost and abrasion resistance are less favorable for this service. Nylon is light and low friction, but it is not the main choice for abrasive dusty limestone service. Hard iron gives the best match to the low-maintenance design goal.

3.2.10. Final Material Selection

Table 3.19. Final Selected Materials

Component	Selected Material	Reason for Selection
Screw flights	AR400 steel	Highest wear resistance and strength for direct abrasive limestone contact
Shaft	1045 steel	Strong, stiff, machinable, and suitable for torque transmission
Trough	Carbon steel	Economical, weldable, and suitable for large fabricated steel construction
Cover	Carbon steel	Low cost and suitable for sheet-metal dust containment
Hanger bearing	Hard iron	Durable wear-resistant material for dusty abrasive support service

The final material selection balances wear resistance, cost, manufacturability, and service life. AR400 is used only where the wear demand is highest. 1045 steel is used for rotating shaft parts that require strength and machining. Carbon steel is used for large fabricated parts because it reduces cost and supports standard manufacturing processes. Hard iron is used for the hanger bearing because it is one of the most exposed support components in the limestone flow environment. This material plan supports the final design goal of a low-maintenance, cost-controlled screw conveyor.

3.3 Manufacturing Process Selection

Manufacturing process selection helps to determine how each component will be produced efficiently, economically, and reliably. Since the conveyor is a large industrial product designed to handle abrasive crushed limestone, the selected processes must support strong steel construction, wear resistance, accurate shaft alignment, and low manufacturing cost. The process selection was based on few factors such as production volume, material type, component size, shape complexity, section thickness for key structural components. For this design, fabrication based methods such as sheet metal forming, welding, and conventional machining are most suitable, while standard parts such as bearings, seals, motor, and gearbox are selected as buy out components to reduce complexity and improve maintainability.

3.3.1. Key Manufacturing Components

The important manufacturing components are:

- Material: Crushed Limestone (Abrasive)
- Configuration: 30" Screw Conveyor, Low RPM
- Capability: Large parts, economical steel
- Requirement: Wear-resistant surfaces

- Production: Low Volume (250 Units/ Year)
- Material: Steel (Carbon)

3.3.2. Economic Batch Size Screening

The production volume is treated as low volume industrial production, around 250 units per year. For this range, the project considered several manufacturing categories, including sand casting, die casting, forging, sheet forming, electro-machining, conventional machining, welding, and polymer/composite processes (Figure 15). Processes such as die casting and electro-machining were screened out because they are too expensive or unsuitable for large steel conveyor components. Polymers and composites were also screened out because the conveyor is a large industrial product carrying abrasive limestone.

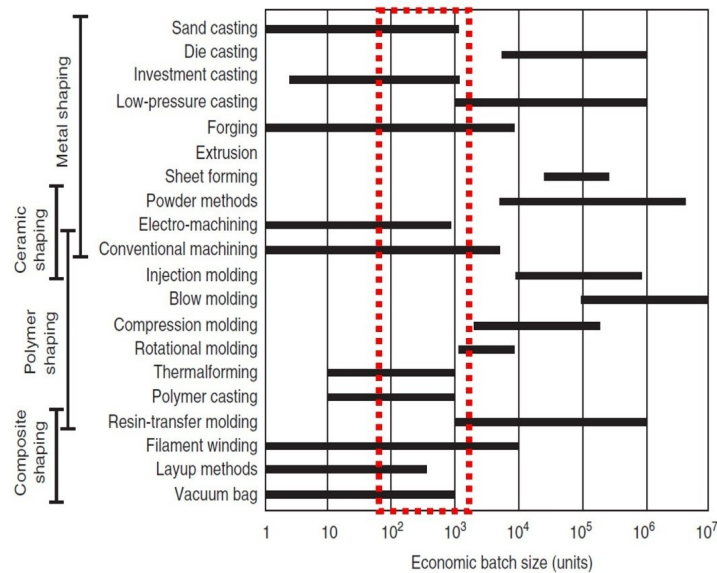


Figure 15. Manufacturing Process for Economic Batch Size.

3.3.3. Shape, Size, and Thickness Screening

The conveyor includes different part shapes (Figure 15). And different part shapes have different suitable manufacturing processes (Figure 16).

Based on Figure 16 and Figure 17, different shape types for the screw conveyor and their manufacturing process is illustrated in the Table 3.19.

Table 3.19. Manufacturing Process for Conveyor based on Shape Classification

Shape Type	Conveyor Parts	Suitable Process
Round/rotational parts	Shafts, pipe, journals	Machining
Flat/plate parts	Covers, brackets, flanges	Sheet metal forming, cutting
Trough/channel parts	U-trough, covers, supports	Sheet metal forming and welding
Helical parts	Screw flights	Formed flight process

Abbreviation	0 Uniform cross section	1 Change at end	2 Change at center	3 Spatial curve	4 Closed one end	5 Closed both ends	6 Transverse element	7 Irregular (complex)
R(ound)								
B(ar)								
S(ection, open) SS(emiclosed)								
T(ube)								
F(lat)								
Sp(herical)								
U(ndercut)								

Figure 16. Shape Classification Selection for Screw Conveyor.

Process	Capability for Producing Shapes
Casting processes	
Sand casting	Can make all shapes
Plaster casting	Can make all shapes
Investment casting	Can make all shapes
Permanent mold	Can make all shapes except T3, T5; F5; U2, U4, U7
Die casting	Same as permanent mold casting
Deformation processes	
Open-die forging	Best for R0 to R3; all B shapes; T1; F0; Sp6
Hot impression die forging	Best for all R, B, and S shapes; T1, T2; Sp
Hot extrusion	All 0 shapes
Cold forging/cold extrusion	Same as hot die forging or extrusion
Shape drawing	All 0 shapes
Shape rolling	All 0 shapes
Sheet-metal working processes	
Blanking	F0 to F2; T7
Bending	R3; B3; S0, S3, S7; T3; F3, F6, F4; S7
Stretching	T4; F4, F7
Deep drawing	T1, T2, T4, T6; F4, F5
Spinning	
Polymer processes	
Extrusion	All 0 shapes
Injection molding	Can make all shapes with proper coring
Compression molding	All shapes except T3, T5, T6, F5, U4
Sheet thermoforming	T4, F4, F7, S5
Powder metallurgy processes	
Cold press and sinter	All shapes except S3, T2, T3, T5, T6, F3, F5, all U shapes
Hot isostatic pressing	All shapes except T5 and F5
Powder injection molding	All shapes except T5, F5, U1, U4
PM forging	Same shape restrictions as cold press and sinter
Machining processes	
Lathe turning	R0, R1, R2, R7; T0, T1, T2; Sp1, Sp6; U1, U2
Drilling	T0, T6
Milling	All B, S, SS shapes; F0 to F4; F6, F7, U7
Grinding	Same as turning and milling
Honing, lapping	R0 to R2; B0 to B2; B7; T0 to T2, T4 to T7; F0 to F2; Sp

Based on data from J. A. Schey, *Introduction to Manufacturing Processes*.

Figure 17. Manufacturing Process based on Shape Classification.

The thickness ranges of trough plates, covers, flights, brackets, and wear liners fall within the practical range for sand casting, sheet forming, conventional machining, and welding (Figure 18). This supports the decision to use fabricated steel construction instead of casting the entire conveyor.

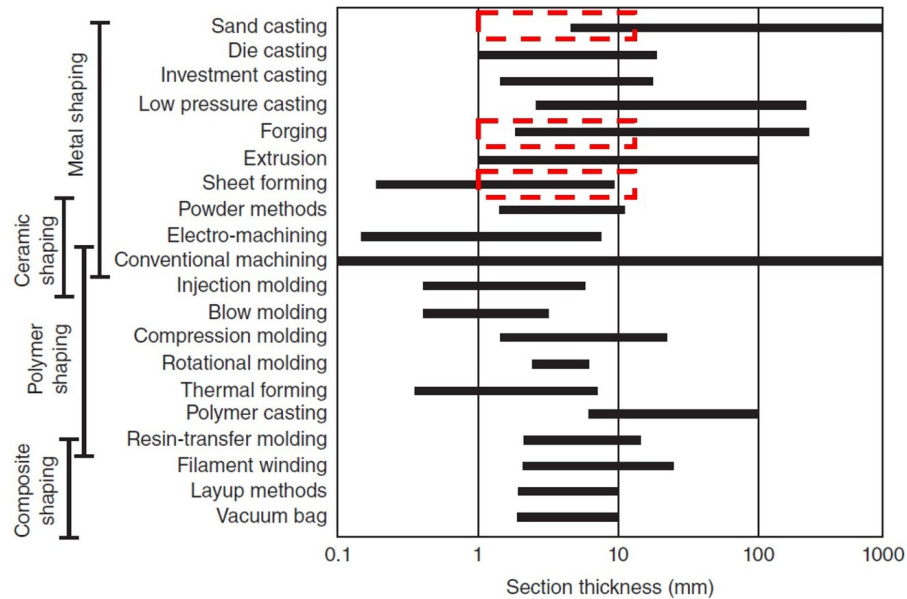


Figure 18. Screening by Section Thickness.

3.3.4. PRIMA Process Selection

The PRIMA selection matrix (Figure 19) shows which materials and manufacturing processes are commonly used together in practice. Based on this matrix, the previously screened processes were confirmed to be suitable for the screw conveyor components and aligned well with the selected materials and design requirements. The selected manufacturing route is mainly a fabrication based process using sheet forming, machining, and welding. This is appropriate because screw conveyors are typically large, steel, low-to-medium volume industrial products.

MATERIAL	IRON	STEEL (carbon)	STEEL (tool, alloy)	STAINLESS STEEL	COPPER & ALLOYS	ALUMINUM & ALLOYS	MAGNESIUM & ALLOYS	ZINC & ALLOYS	TIN & ALLOYS	LEAD & ALLOYS	NICKEL & ALLOYS	TITANIUM & ALLOYS	THERMOPLASTICS	THERMOSETS	FR COMPOSITES	CERAMICS	REFRACTORY METALS	PRECIOUS METALS
QUANTITY																		
VERY LOW 1 TO 100	(1.5) (1.7) (4.6)	(1.5) (1.7) (3.10) (4.6) (5.1) (5.5) (5.6)	(1.1) (1.5) (1.7) (3.10) (4.6) (5.1) (5.5) (5.6)	(1.6) (1.7) (3.7) (3.10) (4.6) (5.1) (5.5) (5.6)	(1.5) (1.7) (3.10) (4.6) (5.1)	(1.5) (1.7) (3.7) (3.10) (4.6) (5.5)	(1.6) (1.7) (3.10) (4.6) (5.5)	(1.1) (1.7) (3.10) (4.6) (5.5)	(1.1) (1.7) (3.10) (4.6) (5.5)	(1.1) (1.7) (3.10) (4.6) (5.5)	(1.1) (1.7) (3.10) (4.6) (5.5)	(1.1) (1.7) (3.10) (4.6) (5.5)	(2.3) (2.7) (3.10) (3.7) (4.6) (5.5) (5.6)	(2.3) (2.7) (3.10) (3.7) (4.6) (5.5) (5.6)	(2.3) (2.7) (3.10) (3.7) (4.6) (5.5) (5.6)	(2.3) (2.7) (3.10) (3.7) (4.6) (5.5) (5.6)	(1.1) (1.7) (3.10) (4.6) (5.5) (5.6)	(1.1) (1.7) (3.10) (4.6) (5.5) (5.6)
LOW 100 TO 1,000	(1.2) (1.5) (1.7) (3.10) (4.6) (5.1) (5.5) (5.6)	(1.2) (1.5) (1.7) (3.10) (4.6) (5.1) (5.5) (5.6)	(1.1) (1.5) (1.7) (3.10) (4.6) (5.1) (5.5) (5.6)	(1.2) (1.5) (1.7) (3.10) (4.6) (5.1) (5.5) (5.6)	(1.2) (1.5) (1.7) (3.10) (4.6) (5.1) (5.5) (5.6)	(1.2) (1.5) (1.7) (3.10) (4.6) (5.1) (5.5) (5.6)	(1.2) (1.5) (1.7) (3.10) (4.6) (5.1) (5.5) (5.6)	(1.1) (1.7) (3.10) (4.6) (5.5)	(1.1) (1.7) (3.10) (4.6) (5.5)	(1.1) (1.7) (3.10) (4.6) (5.5)	(1.1) (1.7) (3.10) (4.6) (5.5)	(1.1) (1.7) (3.10) (4.6) (5.5)	(2.3) (2.7) (3.10) (3.7) (4.6) (5.5) (5.6)	(2.3) (2.7) (3.10) (3.7) (4.6) (5.5) (5.6)	(2.3) (2.7) (3.10) (3.7) (4.6) (5.5) (5.6)	(2.3) (2.7) (3.10) (3.7) (4.6) (5.5) (5.6)	(1.1) (1.7) (3.10) (4.6) (5.5) (5.6)	(1.1) (1.7) (3.10) (4.6) (5.5) (5.6)
LOW TO MEDIUM 1,000 TO 10,000	(1.2) (1.5) (1.7) (3.10) (4.6) (5.1) (5.5) (5.6)	(1.2) (1.5) (1.7) (3.10) (4.6) (5.1) (5.5) (5.6)	(1.1) (1.5) (1.7) (3.10) (4.6) (5.1) (5.5) (5.6)	(1.2) (1.5) (1.7) (3.10) (4.6) (5.1) (5.5) (5.6)	(1.2) (1.5) (1.7) (3.10) (4.6) (5.1) (5.5) (5.6)	(1.2) (1.5) (1.7) (3.10) (4.6) (5.1) (5.5) (5.6)	(1.2) (1.5) (1.7) (3.10) (4.6) (5.1) (5.5) (5.6)	(1.1) (1.7) (3.10) (4.6) (5.5)	(1.1) (1.7) (3.10) (4.6) (5.5)	(1.1) (1.7) (3.10) (4.6) (5.5)	(1.1) (1.7) (3.10) (4.6) (5.5)	(1.1) (1.7) (3.10) (4.6) (5.5)	(2.3) (2.7) (3.10) (3.7) (4.6) (5.5) (5.6)	(2.3) (2.7) (3.10) (3.7) (4.6) (5.5) (5.6)	(2.3) (2.7) (3.10) (3.7) (4.6) (5.5) (5.6)	(2.3) (2.7) (3.10) (3.7) (4.6) (5.5) (5.6)	(1.1) (1.7) (3.10) (4.6) (5.5) (5.6)	(1.1) (1.7) (3.10) (4.6) (5.5) (5.6)
MEDIUM TO HIGH 10,000 TO 100,000	(1.2) (1.5) (1.7) (3.10) (4.6) (5.1) (5.5) (5.6)	(1.2) (1.5) (1.7) (3.10) (4.6) (5.1) (5.5) (5.6)	(1.1) (1.5) (1.7) (3.10) (4.6) (5.1) (5.5) (5.6)	(1.2) (1.5) (1.7) (3.10) (4.6) (5.1) (5.5) (5.6)	(1.2) (1.5) (1.7) (3.10) (4.6) (5.1) (5.5) (5.6)	(1.2) (1.5) (1.7) (3.10) (4.6) (5.1) (5.5) (5.6)	(1.2) (1.5) (1.7) (3.10) (4.6) (5.1) (5.5) (5.6)	(1.1) (1.7) (3.10) (4.6) (5.5)	(1.1) (1.7) (3.10) (4.6) (5.5)	(1.1) (1.7) (3.10) (4.6) (5.5)	(1.1) (1.7) (3.10) (4.6) (5.5)	(1.1) (1.7) (3.10) (4.6) (5.5)	(2.3) (2.7) (3.10) (3.7) (4.6) (5.5) (5.6)	(2.3) (2.7) (3.10) (3.7) (4.6) (5.5) (5.6)	(2.3) (2.7) (3.10) (3.7) (4.6) (5.5) (5.6)	(2.3) (2.7) (3.10) (3.7) (4.6) (5.5) (5.6)	(1.1) (1.7) (3.10) (4.6) (5.5) (5.6)	(1.1) (1.7) (3.10) (4.6) (5.5) (5.6)
HIGH 100,000+	(1.2) (1.5) (1.7) (3.10) (4.6) (5.1) (5.5) (5.6)	(1.2) (1.5) (1.7) (3.10) (4.6) (5.1) (5.5) (5.6)	(1.1) (1.5) (1.7) (3.10) (4.6) (5.1) (5.5) (5.6)	(1.2) (1.5) (1.7) (3.10) (4.6) (5.1) (5.5) (5.6)	(1.2) (1.5) (1.7) (3.10) (4.6) (5.1) (5.5) (5.6)	(1.2) (1.5) (1.7) (3.10) (4.6) (5.1) (5.5) (5.6)	(1.2) (1.5) (1.7) (3.10) (4.6) (5.1) (5.5) (5.6)	(1.1) (1.7) (3.10) (4.6) (5.5)	(1.1) (1.7) (3.10) (4.6) (5.5)	(1.1) (1.7) (3.10) (4.6) (5.5)	(1.1) (1.7) (3.10) (4.6) (5.5)	(1.1) (1.7) (3.10) (4.6) (5.5)	(2.3) (2.7) (3.10) (3.7) (4.6) (5.5) (5.6)	(2.3) (2.7) (3.10) (3.7) (4.6) (5.5) (5.6)	(2.3) (2.7) (3.10) (3.7) (4.6) (5.5) (5.6)	(2.3) (2.7) (3.10) (3.7) (4.6) (5.5) (5.6)	(1.1) (1.7) (3.10) (4.6) (5.5) (5.6)	(1.1) (1.7) (3.10) (4.6) (5.5) (5.6)
ALL QUANTITIES	(1.1)	(1.1) (1.5) (1.7) (3.10) (4.6) (5.1) (5.5) (5.6)	(1.1) (1.5) (1.7) (3.10) (4.6) (5.1) (5.5) (5.6)	(1.1) (1.5) (1.7) (3.10) (4.6) (5.1) (5.5) (5.6)	(1.1) (1.5) (1.7) (3.10) (4.6) (5.1) (5.5) (5.6)	(1.1) (1.5) (1.7) (3.10) (4.6) (5.1) (5.5) (5.6)	(1.1) (1.5) (1.7) (3.10) (4.6) (5.1) (5.5) (5.6)	(1.1) (1.7) (3.10) (4.6) (5.5)	(1.1) (1.7) (3.10) (4.6) (5.5)	(1.1) (1.7) (3.10) (4.6) (5.5)	(1.1) (1.7) (3.10) (4.6) (5.5)	(1.1) (1.7) (3.10) (4.6) (5.5)	(2.3) (2.7) (3.10) (3.7) (4.6) (5.5) (5.6)	(2.3) (2.7) (3.10) (3.7) (4.6) (5.5) (5.6)	(2.3) (2.7) (3.10) (3.7) (4.6) (5.5) (5.6)	(2.3) (2.7) (3.10) (3.7) (4.6) (5.5) (5.6)	(1.1) (1.7) (3.10) (4.6) (5.5) (5.6)	(1.1) (1.7) (3.10) (4.6) (5.5) (5.6)

KEY TO MANUFACTURING PROCESS PRIMA SELECTION MATRIX:

CASTING PROCESSES
(1.1) SAND CASTING
(1.2) GRAVITY DIE CASTING
(1.3) PRESSURE DIE CASTING
(1.4) CENTRIFUGAL CASTING
(1.5) INVESTMENT CASTING
(1.6) CERAMIC MOLD CASTING
(1.7) PLASTER MOLD CASTING
(1.8) SQUEEZE CASTING

PLASTIC & COMPOSITE PROCESSING
(2.1) INJECTION MOULDING
(2.2) REACTION INJECTION MOULDING
(2.3) COMPRESSION MOULDING
(2.4) TRANSFER MOULDING
(2.5) VACUUM MOULDING
(2.6) BLOW MOULDING
(2.7) ROTATIONAL MOULDING
(2.8) CONTACT MOULDING
(2.9) CONTINUOUS EXTRUSION (PLASTICS)

FORMING PROCESSES
(3.1) CLOSED DIE FORGING
(3.2) ROLLING
(3.3) DRAWING
(3.4) COLD FORMING
(3.5) COLD HEADING
(3.6) SWAGING
(3.7) SHEET-METAL FORMING
(3.8) SHEET-METAL SHEARING
(3.9) SHEET-METAL FORMING
(3.10) SPINNING
(3.11) POWDER METALLURGY
(3.12) CONTINUOUS EXTRUSION (METALS)

MACHINING PROCESSES
(4.1) AUTOMATIC MACHINING
(4.2) MANUAL THICKENING
(4.3) LASER BEAM MACHINING (LBM)
(4.4) CHEMICAL MACHINING (CM)
(4.5) ULTRASONIC MACHINING (USM)
(4.6) ABRASIVE JET MACHINING (AJM)

NTM PROCESSES
(5.1) ELECTRICAL DISCHARGE MACHINING (EDM)
(5.2) ELECTROCHEMICAL MACHINING (ECM)
(5.3) ELECTRON BEAM MACHINING (EBM)
(5.4) LASER BEAM MACHINING (LBM)
(5.5) CHEMICAL MACHINING (CM)
(5.6) ULTRASONIC MACHINING (USM)
(5.7) ABRASIVE JET MACHINING (AJM)

(THE ABOVE HEADINGS COVER A BROAD RANGE OF MACHINING PROCESSES AND LEVELS OF CONTROL TECHNOLOGY. FOR MORE DETAIL, THE READER IS REFERRED TO THE INDIVIDUAL PROCESSES.)

Figure 19. PRIMA Selection Matrix.

3.3.5. Recommended Manufacturing Processes for Key Components

The manufacturing plan assigns different processes to different components based on function, shape, and precision requirements.

Table 3.20. Recommended Manufacturing Process

Component	Process	Justification
Screw flight	Forming and welding	– Better fabricated than cast – Favors steel plate for abrasive duty – Standard volume avoids high tooling costs – easy repair
Screw core	Machining and welding	– Tubular shape is naturally sourced – good stiffness to cost balance – easy integration with other parts
End shafts	Machining	– Rotational precision parts require controlled fits – High carbon steel is better than stainless here
Trough	Welding	– Ideal open section for fabrication – economical for large industrial size – easy to add supports/ liners
Covers, flanges, brackets, support frame	Cutting, forming, welding	– Low complexity – easy assembly – low tooling investment – adaptable for design updates
Bearings and standard parts	Buy out	– Improves manufacturability and maintenance by using optimized standard catalog components

This manufacturing plan is well aligned with design for manufacturing principles. It avoids unnecessary high cost processes and uses standard parts where possible.

3.4 Tolerance Stack-up Analysis

This section presents the tolerance and stack-up analysis of the selected screw conveyor design, focusing on critical dimensional chains that influence assembly accuracy, operational performance, and long-term reliability. Two key areas are considered: the conveyor length stack-up (shaft–coupling–bearing alignment) and the screw–trough clearance. The conveyor length stack-up is essential to ensure proper alignment of rotating components, preventing vibration, excessive wear, and premature failure, especially in long assemblies where tolerance accumulation becomes significant. The screw–trough clearance is equally critical, as insufficient clearance can cause rubbing and damage, while excessive clearance reduces conveying efficiency and leads to material loss. These critical dimensions are selected to align with the low-maintenance design objective of Candidate B, ensuring smooth operation, durability, and controlled wear under real manufacturing variations.

3.4.1. Tolerance Analysis of Conveyor Length Stack-up (Shaft–Coupling–Bearing Alignment)

This section presents a detailed tolerance analysis of the conveyor length stack-up associated with the shaft–coupling–bearing alignment of the selected design (Candidate B). The analysis begins with the definition of nominal dimensions and assigned manufacturing tolerances for key components in the assembly. A dimensional chain is then established to evaluate the critical axial gap, followed by both

worst-case and statistical (RSS) tolerance stack-up methods to assess variation. The results are used to verify that no interference occurs under all conditions and to identify dominant contributors to dimensional variation. This provides a basis for targeted design improvement and tighter control of critical dimensions.

3.4.1.1. Selected Nominal Dimensions

The nominal dimensions for the stack-up are selected to represent a realistic configuration for a heavy-duty 30-inch screw conveyor using standard components. These dimensions define the axial relationship between the bearing datum, shaft seats, coupling hub, and spacer. The governing relationship for the assembly gap is expressed as:

where represents the remaining clearance after assembly. Substituting the nominal values results in:

This nominal gap ensures sufficient clearance for assembly and operation.

3.4.1.2. Assigned Tolerances

Balanced bilateral tolerances are assigned to each dimension based on manufacturing processes and functional importance and is summarized in Table 3.21. Larger tolerances are applied to fabricated housing dimensions, while tighter tolerances are used for machined shaft and coupling components.

Table 3.21. Assigned Tolerances

Dimension	Description	Nominal (in)	Tolerance ($\pm$ in)
A	Bearing datum to housing reference face	20.000	0.020
B	Left shaft seat length	5.000	0.005
C	Coupling hub width	7.500	0.010
D	Right shaft spacer/seat length	7.250	0.005

This distribution as shown in Figure 20, reflects practical manufacturing capability and cost considerations.

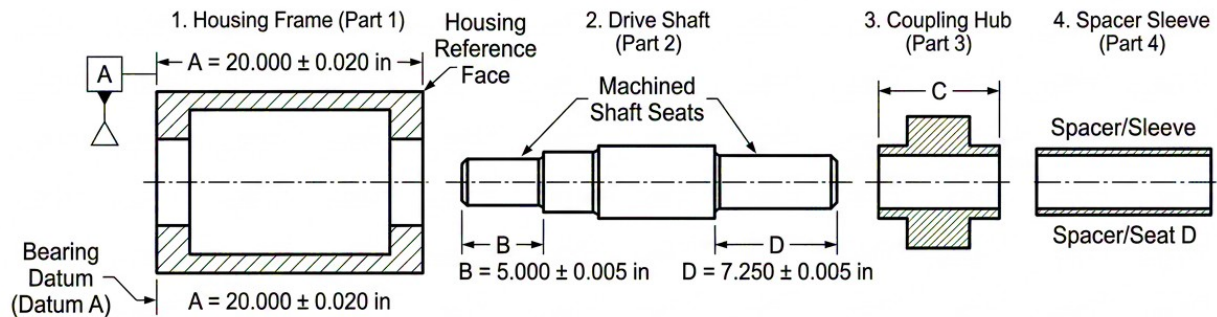


Figure 20. Components with Identified Geometric Features.

3.4.1.3. Tolerance Chain Formulation

A dimensional loop is established along the axial direction of the assembly, linking the housing, shaft, coupling, and spacer components. The tolerance chain (Table 3.22) is evaluated using both direct summation and squared tolerance contributions to support worst-case and statistical analysis and is illustrated in Figure 21.

Table 3.22. Assigned Tolerance Chain

Dimension	Tolerance ($\pm$ in)	Contribution
A	0.020	0.000400
B	0.005	0.000025
C	0.010	0.000100
D	0.005	0.000025
Total	—	0.000550

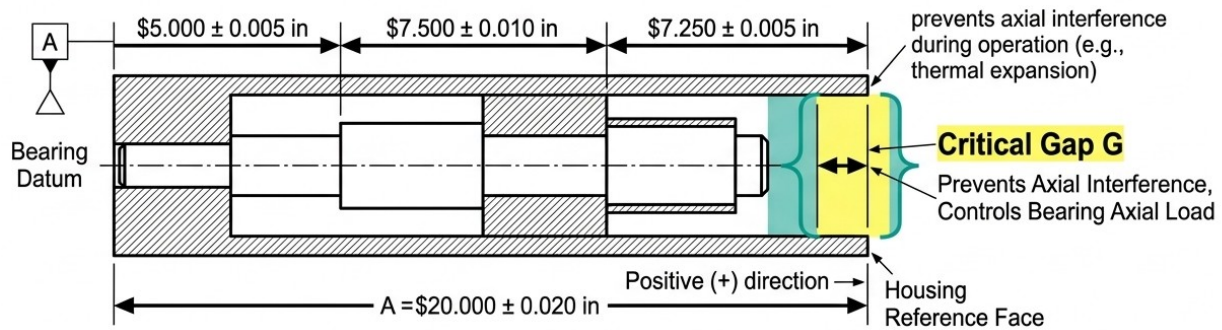


Figure 21. Assembled Drive End Cross-Section with Dimension Loop Analysis.

3.4.1.4. Worst-Case Tolerance Stack-up

The worst-case analysis assumes all tolerances accumulate in the same direction, providing a conservative estimate of variation. The total tolerance is calculated as:

The resulting gap range is:

Since the minimum gap remains positive, the design avoids interference even under the most extreme conditions.

3.4.1.5. Statistical (RSS) Tolerance Analysis

The statistical method assumes independent variation of dimensions and combines tolerances using the root-sum-square (RSS) approach:

This results in a tighter expected variation range:

The statistical result confirms improved consistency while still maintaining a positive minimum gap.

3.4.1.6. Comparison of Results

A comparison between the worst-case and statistical methods highlights the difference between conservative and realistic predictions and is summarized in Table 3.23.

Table 3.23. Comparison of Results

Method	Nominal Gap (in)	Tolerance (in)	Min Gap (in)	Max Gap (in)
Worst-case	0.250	± 0.0400	0.2100	0.2900
Statistical (RSS)	0.250	± 0.02345	0.2266	0.2735

Both approaches confirm that the assembly remains acceptable, as the gap stays positive in all cases.

3.4.1.7. Sensitivity / Tolerance Contribution Analysis

The contribution of each dimension to the total variation is evaluated based on squared tolerances. The percentage contributions are summarized in Table 3.24.

Table 3.24. Tolerance Contribution

Dimension	Contribution (%)
A	72.73
B	4.55
C	18.18
D	4.55

The results show that dimension A (bearing datum–housing reference distance) dominates the variation, accounting for nearly three-quarters of the total.

3.4.1.8. Design Recommendation

Since dimension A is the primary driver of variation, improving gap control should focus on tightening its tolerance. Reducing variability in the housing-related dimension will significantly improve overall assembly consistency without requiring tighter control on other components, which contribute minimally. This targeted approach ensures better performance while maintaining cost-effective manufacturing.

3.4.2. Tolerance Analysis of Screw–Trough Clearance

This section evaluates the radial clearance between the screw outside diameter and the trough inside diameter. This clearance is critical because insufficient clearance can cause rubbing, wear, and mechanical damage, while excessive clearance can reduce conveying efficiency and increase material loss. The analysis verifies that the selected 30 in screw design maintains acceptable clearance under realistic manufacturing variation.

3.4.2.1. Critical Dimension

The critical dimension for this analysis is the radial screw–trough clearance. It is calculated from the trough inside diameter and screw outside diameter as the following equation where C is the radial screw–trough clearance, T is the trough inside diameter, and S is the screw outside diameter. The value is divided by two because the total diametral clearance is distributed on both sides of the screw.

3.4.2.2. Assumed Nominal Dimensions

For the selected heavy-duty 30 in screw conveyor, the trough inside diameter is assumed to be 30.500 in and the screw outside diameter is 30.000 in. The nominal radial clearance is:

3.4.2.3. Assigned Bilateral Tolerances

Realistic bilateral tolerances are assigned to the trough inside diameter and screw outside diameter (Table 3.25). Since the clearance equation depends on both dimensions, each tolerance contributes to the final clearance variation.

Table 3.25. Assigned Tolerance

Dimension	Description	Nominal Dimension (in)	Tolerance ($\pm$ in)
(T)	Trough inside diameter	30.500	0.030
(S)	Screw outside diameter	30.000	0.020

Because the clearance equation is $C = (T - S) / 2$, the sensitivity factor for the trough inside diameter is $1/2$, while the sensitivity factor for the screw outside diameter is $-1/2$.

3.4.2.4. Worst-Case Stack-up

The worst-case clearance tolerance is calculated by combining the two dimensional tolerances and dividing by two:

Therefore, the screw–trough clearance becomes:

The minimum clearance remains positive, so the screw does not interfere with the trough under worst-case tolerance conditions.

3.4.2.5. Statistical RSS Stack-up

The statistical tolerance stack-up (Figure 22) is calculated using the root-sum-square method with the sensitivity factors included.

Thus, the statistical clearance result is:

This confirms that the clearance remains acceptable under statistical tolerance variation.

3.4.2.6. Sensitivity and Contribution Analysis

The squared tolerance contributions are used to determine which dimension contributes most to clearance variation.

Table 3.26. Sensitivity Calculation

Dimension	Squared Contribution	Percentage Contribution
Trough inside diameter (T)	0.000225	69.23%
Screw outside diameter (S)	0.000100	30.77%
Total	0.000325	100%

The trough inside diameter is the largest contributor to clearance variation at 69.23%. Therefore, the most effective way to improve screw–trough clearance control is to tighten the trough inside diameter tolerance. This adjustment improves clearance consistency while directly supporting the low-maintenance objective of the selected design.

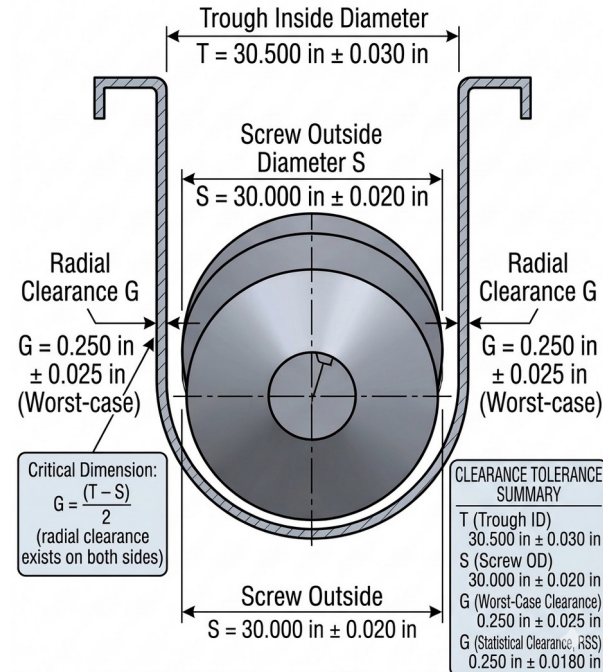


Figure 22. RSS / Statistical Stack-up of Screw–Trough Clearance.

3.5 DFM Analysis

The design was evaluated using Design for Manufacturing (DFM) principles to assess the ease of fabrication and identify potential manufacturing challenges. The analysis focused on both sheet metal components and machined components.

For primary structural components such as the screw flight, trough, and covers, the evaluation considered factors including bend angles, material grain direction, bend radius, and geometric proportions. The results indicate that most components are compatible with standard sheet metal forming operations, with acceptable bend conditions and uniform thickness distribution.

Table 3.27. DFM Analysis (Structural Components)

	1	2	3	4		
PART(ID or Description or features)	Hanger U-Shape	Hanger Bolt	Shaft	Trough End		
Step-1(Part Details)						
Quantity	1	2	1	2		
Primary operation on the part	Mill	Turn / Thread	Turn	Mill		
Secondary operation on the part	Drill / Finish	Finish	Keyway / Finish	Drill / Finish		
Instances of tolerances	2	2	3	2		
Step-2(DFM analysis)						
					Sum	
Number of secondary operations on the part (+1 for each operation)	1	1	1	1	4	
Shape of the machined part(+1 if	-1	0	1	0	0	

taper, +1 if complex contour,-1 if rectangular)						
Can the part withstand forces of clamping? (-1 if yes and +1 if no)	-1	-1	-1	-1	-4	
Presence of secure holding and fixturing (-1 if yes and +1 if no)	-1	-1	-1	-1	-4	
Number of sharp corners(+1 for each occurrence)	1	0	0	0	1	
Presence of stock dimensions(-2 if present)	-2	-2	0	0	-4	
Presence of undercuts (+1 for each occurrence)	0	0	0	0	0	
Presence of parting lines or draft surface for clamping (+1 for each occurrence)	0	0	0	0	0	
Presence of shoulders and overruns (+1 for each occurrence)	0	0	1	0	1	
Type of cutter (+1 if small and -1 if large)	1	0	0	1	2	
Enough access for cutters, bushings and fixture elements	-1	-1	-1	-1	-4	
Interrupted cuts for single point cutting tools (+1 for each occurrence)	1	1	1	0	3	
Space for burr relief (-1 if present or +1 if not present)	-1	-1	-1	-1	-4	
				Total	-9	(0 is ideal)
Step-3. Miscellaneous checklist						
Presence of finishing stock(+1 if less than 0.15 in)	0	0	0	0	0	
Can quantity of parts be reduced? (-1 for each quantity reduced)	0	0	0	0	0	
Can the instances of tolerances be reduced?(-1 for each occurrence reduced)	0	0	0	0	0	
Is the material difficult to machine? (+1 if yes)	1	0	0	1	2	
Is the cutting tool standard? (-1 if yes and +1 if no)	-1	-1	-1	-1	-4	
Total	-3	-5	-1	-2	-2	(0 is ideal)

The screw flight exhibits a higher number of tolerance requirements due to its geometric complexity; however, it remains manufacturable using cold forming processes. The overall DFM score for structural components indicates acceptable manufacturability with opportunities for minor improvements.

For hanger components, which involve machining operations, the analysis focused on tool accessibility, fixturing, number of secondary operations, and geometric complexity. The results show that the

components are well-suited for conventional machining processes such as turning and milling. The use of standard stock dimensions and minimal undercuts contributes to efficient manufacturing.

Table 3.28. DFM Analysis (Hanger Components)

	1	2	3		
PART(ID or Description or features)	Screw Shaft/Pipe	Coupling/Drive Shaft	Trough Body		
Step-1(Part Details)					
Quantity	1	5	1		
Primary operation on the part	Sand Casting	Sand Casting	Sand Casting		
Secondary operation on the part	Machining / finishing	Machining / finishing	Trimming / finishing		
Instances of tolerances	3	3	4		
Step-2(DFM analysis)					
				Sum	
Number of secondary operations on the part (+1 for each operation)	2	2	1	5	
Section thickness of the casting must be uniform(-1 if yes and +1 if no)	-1	-1	-1	-3	
Straight parting lines (-1 if yes); Stepped parting lines(+1 if yes)	-1	-1	-1	-3	
Elimination of an undercut on the casting if present(-1 if yes and +1 if no)	0	0	0	0	
Presence of draft or taper (-1 if yes and +1 if no)	1	1	-1	1	
Presence of riser near heaviest section(-1 if yes and +1 if no)	-1	-1	-1	-3	
Presence of circular web to connect the ribs(-1 if present)	0	0	0	0	
Presence of sharp corners (+1 if yes); Presence of rounded corners(-1 if yes)	-1	-1	1	-1	
Presence of fillets or tapers (-1 if yes); Presence of sharp steps(+1 if yes)	-1	0	0	-1	
Intersection of two walls at right angle is preferable(-1 if yes)	0	0	-1	-1	
use of filter to reduce inclusions (-1 if yes or +1 if no)	-1	-1	-1	-3	
Small cored holes are avoided (+1 if any)	1	1	1	3	
			Total	-6	(0 is ideal)

Step-3. Miscellaneous checklist					
Can number of cores be reduced?(-1 for each quantity reduced)				0	
Can the instances of tolerances be reduced?(-1 for each occurrence reduced)				0	
Is the material difficult to cast ? (+1 if yes)				0	
Total	-2	-1	-3	0	(0 is ideal)

Overall, the DFM evaluation suggests that the design achieves a good balance between performance requirements and manufacturability, with potential improvements in reducing tolerance instances and simplifying certain geometries.

3.6 DFA Analysis

A Design for Assembly (DFA) analysis was conducted to evaluate the efficiency, simplicity, and practicality of assembling the screw conveyor system, with the objective of reducing assembly time, minimizing errors, and improving overall manufacturability. The analysis focused on key factors including part count, ease of handling, insertion and fastening difficulty, alignment requirements, accessibility during assembly, and the potential for standardization of components. Each component was assessed in terms of how easily it can be manipulated, positioned, and secured during assembly, particularly considering the size and weight of heavy-duty conveyor elements.

The results indicate that the assembly process is moderately complex due to the large and heavy components, such as the screw shaft, flights, and trough sections, which require careful handling and precise alignment. Operations such as welding of flights to the shaft and alignment of rotating components demand skilled labor and appropriate fixturing to ensure dimensional accuracy and prevent misalignment issues that could lead to operational failures. Additionally, components like bearings, seals, and couplings require accurate positioning and proper seating, increasing the importance of controlled assembly procedures.

However, the design also incorporates several features that support efficient assembly. The use of modular subassemblies, such as pre-fabricated trough sections and standardized bearing housings, reduces assembly complexity and allows for easier installation and maintenance. The use of bolted connections in non-critical areas improves accessibility and simplifies disassembly for inspection or repair. Furthermore, the number of unique fasteners and components has been minimized where possible to reduce part variety and assembly errors.

Overall, the DFA analysis suggests that while the system inherently involves some assembly challenges due to its scale and functional requirements, opportunities exist to further improve assembly efficiency through better alignment features, increased use of standardized components, and enhanced fixturing methods. These improvements can lead to reduced assembly time, lower labor costs, and improved reliability of the final assembled system.

Table 3.29. DFA Worksheet and Assembly Index Calculation

Part	Retrieve	Handle	Insert	# Parts (P)	Assembly Index	Part Required (PR)
Description or ID	Small < 12 mm +1 < 2 mm +2 Tangled +2 Flexible +2	No End Sym +2 or +1 if obvious No Insert Sym +2 Heavy or Tools Needed +2	Alignment Difficult +2 Obstructed +2 Not Top Down +2 Resistance Hold in Place +1 +3	Enter number of part instances	Sum all assembly scores	Enter code (1,2,3)
Flight Screw	0	1	5	1	7	1,3
Pipe	0	1	4	1	6	1,3
Couplings	0	1	6	3	10	1
End Shaft	0	1	7	1	9	1
Drive Shaft	0	1	7	1	9	1
Trough	0	2	8	4	14	3
Trough Covers	0	2	4	4	10	2
Trough Ends	0	2	2	2	6	3
Hangers	0	1	4	3	8	2
Hanger Bearing	0	1	2	3	6	1
End Pedestal Bearing	0	1	4	2	7	1
Conveyor Inlet Flange	0	2	3	1	6	3
Standard	0	2	5	1	8	3

Discharge Spout						
Bolts	1	2	5	80	88	2,3
Motor Assembly	0	1	8	1	10	1,3
Shaft Seals	0	1	6	2	9	2,3
Total				110	213	11

$$\begin{aligned} \text{Assembly Index} &= 213 \quad (0 \text{ Ideal}) \\ \text{Part Reduction} &= (\#P - \#PR) / \#P = 0.9 \quad (0 \text{ Ideal}) \end{aligned}$$

Part required if:

- 1 relative motion required
- 2 different material required
- 3 assembly/disassembly impossible without it

The system consists of 110 total part instances, resulting in an assembly index of 213, which is significantly higher than the ideal value. This indicates that the current design requires substantial effort in handling and assembly operations. Additionally, the part reduction ratio suggests that several components may be redundant or could be integrated. Fasteners contribute significantly to assembly complexity, both in terms of handling and insertion effort. Based on the DFA analysis, several improvements are proposed to enhance assembly efficiency. The high number of fasteners can be reduced by replacing bolted connections with welded joints in non-serviceable regions. This would significantly reduce assembly time and handling complexity. Modular subassemblies can be introduced, particularly in the hanger and support components, to simplify installation and maintenance. Additionally, the use of snap-fit or simplified fastening mechanisms for non-load-bearing components can further reduce assembly effort.

By implementing these changes, the assembly index can be reduced, leading to improved manufacturability, reduced labor cost, and enhanced overall system reliability.

4. Final Design

This section presents the finalized screw conveyor design after completing refinement, optimization, and tolerance analyses. The selected configuration integrates optimized dimensions, validated clearances, and practical manufacturing considerations to ensure reliable and low-maintenance operation. In addition to dimensional verification, this section evaluates key engineering aspects including safety factors, reliability, cost considerations, and potential failure modes through an FMEA approach. The final design is further supported by detailed CAD models that reflect the complete assembly and component-level decisions. Together, these elements confirm that the design is not only functionally robust but also manufacturable, cost-effective, and suitable for long-term operation.

4.1 Safety Factor Analysis

The safety factor is used to ensure that each component can safely withstand loads beyond expected operating conditions. For this project, safety factors are applied to each critical component of the screw conveyor to ensure reliable operation under abrasive and continuous loading conditions. They account for uncertainties such as material behavior, possible overload, and manufacturing variations. This helps prevent failure and ensures the conveyor performs safely and consistently throughout its service life.

The critical components considered are:

- Flighting
- Screw shaft / end shaft
- Coupling and drive connection
- Hanger bearing
- Trough
- Cover and seals

Bolts, end bearings, motors, and reducer are not evaluated with custom safety factors because they are standard purchased components.

The overall safety factor for each component is determined by combining multiple contributing factors, as shown in the following equation.

The margin of safety indicates how much additional strength a component has beyond the required load. It helps assess whether the design is safe and how close it is to failure. The Margin of Safety for all the critical components is also calculated using the following equation.

$$\text{Margin of Safety} = \text{Overall Safety Factor} - 1$$

The safety factor and the Margin of Safety of safety for each critical component is presented below.

Table 4.1. Overall Safety Factor for Flighting

Safety Factor Type	Safety Factor	Reasoning
Material	1.1	Materials properties known from handbook
Stress	1.2	Potential overload conditions (20- 50%)
Tolerance	1.0	Manufacturing tolerances average
Failure theory	1.1	Failure Analysis based on static uniaxial or multiaxial state of stress
Reliability	1.2	Ensures part reliability 92-98%

Overall Safety Factor = $1.1 \times 1.2 \times 1.0 \times 1.1 \times 1.2 = 1.7424$
Margin of Safety = Overall Safety Factor - 1 = $1.7424 - 1 = 0.7424$

Table 4.2. Overall Safety Factor for Screw Shaft/ End Shaft

Safety Factor Type	Safety Factor	Reasoning
Material	1.1	Materials properties known from handbook
Stress	1.2	Potential overload conditions (20- 50%)
Tolerance	1.0	Tolerances are tight and well held
Failure theory	1.1	Assumes static and fatigue loading conditions.
Reliability	1.3	Ensures part reliability 92-98%
Overall Safety Factor = $1.1 \times 1.2 \times 1.0 \times 1.1 \times 1.2 = 1.8876$		
Margin of Safety = Overall Safety Factor - 1 = $1.8876 - 1 = 0.8876$		

Table 4.3. Overall Safety Factor for Coupling and Drive Connection

Safety Factor Type	Safety Factor	Reasoning
Material	1.1	Materials properties known from handbook
Stress	1.2	Potential overload conditions (20- 50%)
Tolerance	1.0	Tolerances are tight and well held
Failure theory	1.1	Assumes static and fatigue loading conditions.
Reliability	1.5	Ensures part reliability >99%
Overall Safety Factor = $1.1 \times 1.2 \times 1.0 \times 1.1 \times 1.5 = 2.178$		
Margin of Safety = Overall Safety Factor - 1 = $2.178 - 1 = 1.178$		

Table 4.4. Overall Safety Factor for Hanger Bearing

Safety Factor Type	Safety Factor	Reasoning
Material	1.1	Materials properties known from handbook
Stress	1.0	Load is well defined
Tolerance	1.0	Tolerances are tight
Failure theory	1.0	Assumes static and fatigue loading conditions
Reliability	1.1	Ensures part reliability 92-98%
Overall Safety Factor = $1.1 \times 1.0 \times 1.0 \times 1.0 \times 1.1 = 1.21$		
Margin of Safety = Overall Safety Factor - 1 = $1.21 - 1 = 0.21$		

Table 4.5. Overall Safety Factor for Trough

Safety Factor Type	Safety Factor	Reasoning
Material	1.1	Materials properties known from handbook
Stress	1.3	Potential overload conditions (20- 50%)
Tolerance	1.1	Manufacturing Tolerances average
Failure theory	1.1	For static and fatigue loading conditions.

Reliability	1.3	Ensures 92-98% system reliability over time.
Overall Safety Factor = $1.1 \times 1.3 \times 1.1 \times 1.3 = 2.25$		
Margin of Safety = Overall Safety Factor - 1 = $2.25 - 1 = 1.25$		

Table 4.6. Overall Safety Factor for Cover and Seals

Safety Factor Type	Safety Factor	Reasoning
Material	1.1	Materials properties known from handbook
Stress	1.2	Potential overloading (20%)
Tolerance	1.0	Tight Tolerances
Failure theory	1.3	Failure Analysis not well developed
Reliability	1.4	Ensures >99% system reliability.
Overall Safety Factor = $1.1 \times 1.2 \times 1.0 \times 1.3 \times 1.4 = 2.40$		
Margin of Safety = Overall Safety Factor - 1 = $2.40 - 1 = 1.40$		

All calculated overall Safety Factors and Margin of Safety for the components are compiled and presented in Table 4.7.

Table 4.7. Summary of Safety Factor Analysis of Critical Components

Component	Overall Safety Factor	Margin of Safety
Flighting	1.7424	0.7424
Screw shaft / End shaft	1.8876	0.8876
Coupling and drive connection	2.178	1.178
Hanger bearing	1.21	0.21
Trough	2.25	1.25
Cover and seals	2.40	1.40

4.2 Reliability Analysis

Reliability analysis is one of the most important sections of the report. It evaluates whether the conveyor can perform its intended function without failure under expected operating conditions.

4.2.1. Reliability Model and Failure Rates

The reliability analysis uses a constant failure rate model assuming that component failure rates remain approximately constant during the operating period considered. Reliability is calculated as a function of time, failure rate, and operating duration.

The general reliability equation for a constant failure-rate model is:

Where, $R(t)$ is Reliability at time t , λ is Constant failure rate & t is Operating time.

The analysis assumes the component life values are estimated using references and the failure rates are either assumed or obtained from textbooks or reliable sources.

The failure rate table (Table 4.8) identifies which components are more likely to fail and why.

Table 4.8. Failure Rate for Each Component

Component	Failure Rate λ (per 1000 hr)	Reason
Flighting	1×10^{-6}	Wear-dominated, but AR400 provides high abrasion resistance
Screw shaft / end shaft	2×10^{-7}	Strong, low-speed rotating shaft
Coupling and drive connection	1×10^{-6}	Moderate risk due to torque transfer and alignment
Hanger bearing assembly	5×10^{-4}	Most failure-prone internal part due to dust and wear
Trough	5×10^{-7}	Low wear/ corrosion failure rate
Cover and seals	1×10^{-4}	Seal degradation and leakage risk
End bearings	4×10^{-7}	Standard rolling bearings
Motor, reducer, V-belt	1.5×10^{-4}	Standard bought out drive components

The most critical reliability concern is the hanger bearing assembly, which has the highest failure rate. This is expected because it operates inside or near the material flow path and is exposed to dust, abrasion, misalignment, and lubrication challenges. The second important concern is the motor/reducer/V-belt group, followed by cover and seals. These components are not necessarily weak, but they are service sensitive and require regular inspection.

4.2.2. Reliability of the Chosen Design

The reliability values were calculated for each major component based on its assumed life and operating time and illustrated in the table (Table 4.9)

Table 4.9. Calculated Reliability

Component	Life (t)	Reliability
Flighting	10 years	99.991%
Screw shaft/ end shaft	9 years	99.998%
Coupling and drive connection	9 years	99.992%
Hanger bearing	7 years	90.4%
Trough	11 years	99.995%
Cover and seals	3 years	99.74%
End bearings	6 years	99.998%
Motor, reducer, V-belt	11 years	98.56%

Most components have very high reliability values, but the hanger bearing stands out with a much lower reliability of 90.4%. This confirms that hanger bearing design, protection, and maintenance are essential to the overall system reliability. The motor/reducer/V-belt system also has lower reliability than most structural components, at 98.56%. This is reasonable because drive systems are active mechanical components and are more vulnerable to overload, belt slip, misalignment, and wear.

4.2.3. System Reliability Analysis

A system reliability diagram is a visual model that shows how different components are connected and how their failures affect the overall system. The Screw Conveyor system reliability diagram combines series and parallel reliability relationships (Figure 23). Series connections are used for components that must all function for the conveyor to operate, so failure of any one of them can interrupt the system. Parallel connections are used where backup or secondary components are included, allowing the system to continue operating even if one element fails. This approach is used because the screw conveyor does not depend on all parts in the same way. Some components are essential single path elements, while others have supporting or redundant features that improve reliability. By combining both relationships, the analysis gives a more realistic estimate of the overall system reliability.

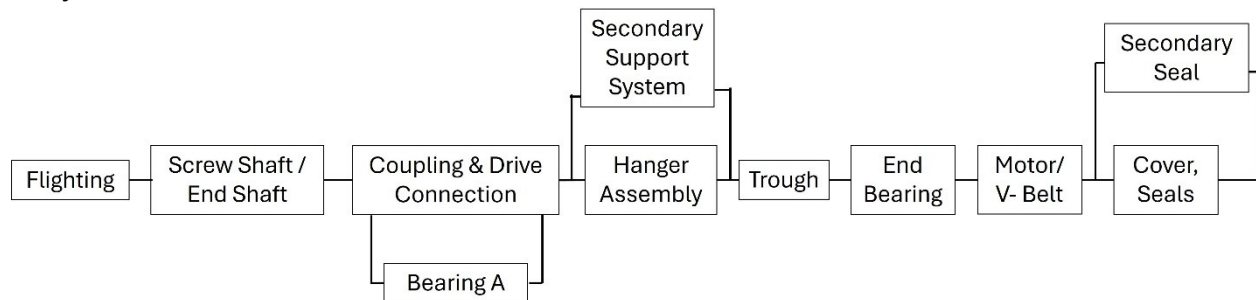


Figure 23. System Reliability Diagram.

For series connection, all components must work for the system to work. So, the system reliability is found by multiplying the reliabilities,

For parallel connection, the system works as long as at least one component works. So, the calculation is done by calculating the chance that all fail, then subtract from 1,

Figure 10 illustrates the system reliability diagram after inserting the calculated reliability values of each screw conveyor component. It shows how the individual component reliabilities are combined through series and parallel relationships to determine the overall reliability of the system. For the secondary parallel branches, lower reliability values were assumed because these components act as backup support elements and are only required to function if the primary component fails.

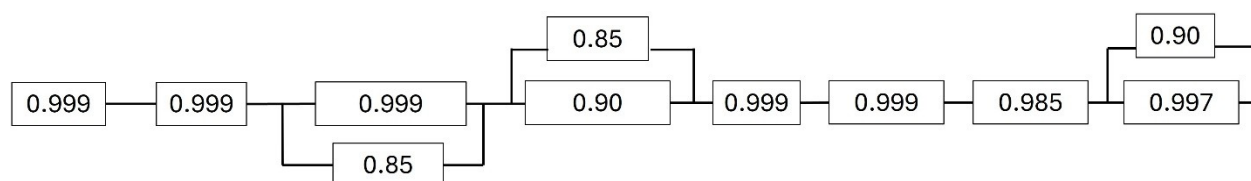


Figure 24. System Reliability Diagram with calculated Reliability.

This yields the system reliability for the screw conveyor,

The resulting system reliability is:

This means the overall system reliability is approximately:

Overall system reliability is calculated as 97.11%, which indicates that the chosen system has a strong reliability level. This design is strongly reliable for a mechanical conveying system operating in an abrasive, dusty environment.

4.2.4. Ways to Improve Reliability

The system's reliability is still influenced heavily by the lowest reliability components, especially the hanger bearing and drive system. Reliability can be improved further by implementing additional preventive measures during design, manufacturing, installation, and maintenance.

Use High-Quality Wear-Resistant Materials: Abrasion-resistant materials such as AR400 should be used in high wear areas such as screw flights and wear zones. This reduces erosion and extends service life.

Optimize Hanger Use: The design should minimize hanger bearings where possible. Hanger bearings create internal obstruction and are exposed to dust and abrasive material. As hanger is necessary, they should be designed with strong sealing, heavy duty bearing materials, and easy maintenance access.

Maintain Precise Alignment: Axial alignment, shaft straightness, coupling alignment, and screw trough clearance must be controlled. Misalignment can cause vibration, rubbing, bearing overload, and premature wear.

Protect Bearings and Seals: Bearings and seals should be protected using dust-tight covers, robust sealing arrangements, and routine inspections. Since dust contamination is a major reliability risk, seal design is directly connected to bearing life.

Simplify Assembly: Modular subassemblies should be used to reduce assembly errors. A simpler assembly process improves reliability because fewer alignment mistakes occur during installation and maintenance.

4.2.5. Mean Time to Failure

Mean Time to Failure, or MTTF, represents the average expected life of a part before it fails.

In this project, MTTF is used as an approximate lifetime measure, although the screw conveyor is actually a repairable system.

Table 4.10. Approximate Life of Each Component for MTTF

Component	Approximate Life
Flighting / AR400	10 years
Screw shaft / end shaft	9 years
Coupling and drive connection	9 years

Hanger bearing	7 years
Trough	11 years
Cover and seals	3 years
End bearings	6 years
Motor, reducer, V-belt	11 years

The calculated MTTF of the screw conveyor assembly is 8.25 years, which indicates a reasonable expected service life for the system. Components such as the trough, motor, reducer, and V-belt show longer life, while cover and seals have the shortest life and may require more frequent inspection or replacement. Overall, the result shows that the conveyor can provide reliable operation if proper maintenance is performed on the lower-life components.

4.2.6. Fault Tree Analysis

The fault tree analysis identifies the possible causes that can lead to the top event, which is the loss of conveying function in the screw conveyor system (Figure 25).

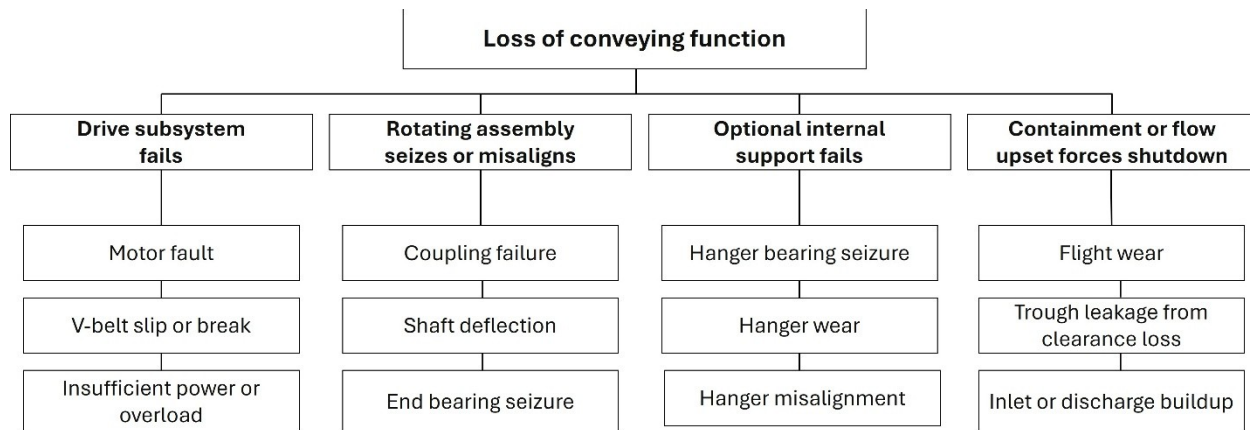


Figure 25. Fault Tree Analysis.

This fault tree is important because it shows the main areas that require design attention to improve reliability. The analysis indicates that proper drive protection, accurate shaft alignment, reliable hanger bearing performance, and controlled material flow are essential for preventing conveyor shutdown. By identifying these failure paths early, suitable preventive actions such as regular inspection, proper lubrication, dust sealing, and timely replacement of worn parts can be planned.

4.3 Failure Modes and Effects Analysis

This section presents a Failure Modes and Effects Analysis (FMEA) for the finalized screw conveyor design. The purpose of this analysis is to systematically identify potential failure modes, evaluate their causes and effects on system performance, and prioritize associated risks. Each failure mode is assessed using occurrence (O), severity (S), and detection (D) ratings on a scale of 1–10, and the Risk Priority Number (RPN) is calculated as:

This approach enables the identification of critical components and operational conditions that may impact reliability, safety, and maintenance requirements. The results are used to guide design improvements, material selection, and operational strategies to ensure a robust and low-maintenance system.

4.3.1. FMEA Evaluation and Risk Assessment

The FMEA was conducted across all major components and processes of the screw conveyor, including the screw flights, shaft, coupling, drive unit, trough, bearings, seals, and material handling sections. The analysis reveals that the highest-risk items are associated with material handling and wear-related failures. In particular, non-uniform feeding or overloading in the inlet section exhibits the highest RPN (200), indicating a significant risk of blockage, increased wear, and reduced system life. Similarly, excessive wear of screw flights (RPN = 192) is identified as a critical concern due to the abrasive nature of limestone, directly affecting conveying capacity and maintenance frequency.

Structural and alignment-related issues also present notable risks. Shaft deflection and hanger bearing wear (both with RPN = 160) can lead to vibration, misalignment, and premature failure if not properly controlled. Trough wear and assembly misalignment further contribute to operational inefficiencies and increased maintenance requirements. Moderate-risk components include the drive unit (RPN = 108) and shaft seals (RPN = 120), which primarily impact system reliability and environmental containment. To mitigate these risks, several corrective actions have been incorporated into the final design. These include the use of wear-resistant materials such as AR400 for screw flights, proper shaft sizing based on torque and deflection analysis, minimizing the number of hanger bearings, and ensuring precise alignment during assembly. Additionally, controlled feeding mechanisms, improved sealing arrangements, and regular inspection and maintenance practices are recommended to further reduce failure likelihood. Overall, the FMEA confirms that while several high-risk areas exist, appropriate design and operational controls have been implemented to manage these risks effectively, resulting in a robust and reliable screw conveyor system. The FMEA was conducted across all major components and processes of the screw conveyor system. The results are summarized in Table 4.11.

Table 4.11. Failure Modes and Effects Analysis

Part	Function	Failure Mode	Effects of Failure	Causes of Failure	O	S	D	RPN	Recommended Corrective Action
Screw flights	Convey crushed limestone along trough	Excessive wear, erosion	Reduced capacity, unstable flow, frequent shutdown	Abrasive material, insufficient wear resistance	6	8	4	192	Use wear-resistant material (AR400), apply liners/hardfacing, periodic inspection
Screw shaft	Transmit torque and support rotation	Shaft bending, excessive deflection	Vibration, rubbing, premature failure	Undersized shaft, long unsupported span	4	8	5	160	Verify shaft design via torque/deflection analysis, reduce span, improve support
Coupling	Connect drive shaft to screw shaft	Misalignment or loosening	Power loss, vibration, internal damage	Poor assembly, weak fastening	4	7	4	112	Ensure alignment, use stronger coupling/bolts, inspect during assembly
Drive unit / motor	Supply rotational power	Motor overload or burnout	Conveyor stoppage, production loss	Incorrect sizing, excessive load	4	9	3	108	Confirm motor sizing with safety factor, use overload protection
Trough	Contain and guide material	Wall wear or perforation	Leakage, dust release, contamination	Abrasion, insufficient thickness	5	8	4	160	Increase thickness, use liners, select wear-resistant material

Trough cover	Limit dust escape	Leakage or looseness	Dust emission, unsafe environment	Poor sealing, loose fasteners	5	7	4	140	Improve sealing, use better gaskets, regular inspection
Hanger bearing	Support intermediate shaft span	Wear or seizure	Increased power draw, misalignment, downtime	Dust ingress, poor lubrication	4	8	5	160	Minimize number of hangers, use sealed bearings, improve lubrication
End bearings	Support shaft ends	Bearing failure	Vibration, stoppage	Overload, contamination, misalignment	4	8	4	128	Select appropriate bearings, ensure proper alignment and sealing
Shaft seals	Prevent leakage at shaft exits	Seal wear or leakage	Material leakage, contamination	Abrasion, improper fit	5	6	4	120	Use abrasion-resistant seals, improve installation and maintenance
Feed / inlet	Introduce material uniformly	Non-uniform feeding, overloading	Blockage, increased wear	Poor feed control, uneven loading	5	8	5	200	Use controlled feed system, ensure uniform loading
Discharge	Release material efficiently	Blockage or buildup	Reduced efficiency, backflow	Poor geometry, sticky material	4	7	5	140	Improve discharge design, allow clean-out access
Assembly / alignment	Ensure correct installation	Misalignment during assembly	Vibration, rubbing, reduced life	Poor tolerance control, improper installation	5	8	4	160	Use alignment checks, verify tolerances before commissioning

4.4 Cost Analysis

This section presents a cost analysis of the finalized screw conveyor design, with the objective of evaluating its economic feasibility alongside its technical performance. The total unit cost is determined by systematically accounting for all major cost components, including material, labor, equipment, overhead, and tooling costs. As illustrated in Figure 26, the cost model follows a structured approach in which each component is calculated individually and then combined to obtain the overall unit cost:

Here, material cost reflects the raw materials used in fabrication, labor cost is based on manufacturing time and labor rates, equipment cost accounts for machine usage, overhead cost is estimated as a percentage of direct costs, and tooling cost is distributed over the number of units produced. This framework ensures a comprehensive and realistic estimation of the total cost, enabling informed design decisions that balance performance, manufacturability, and economic efficiency.

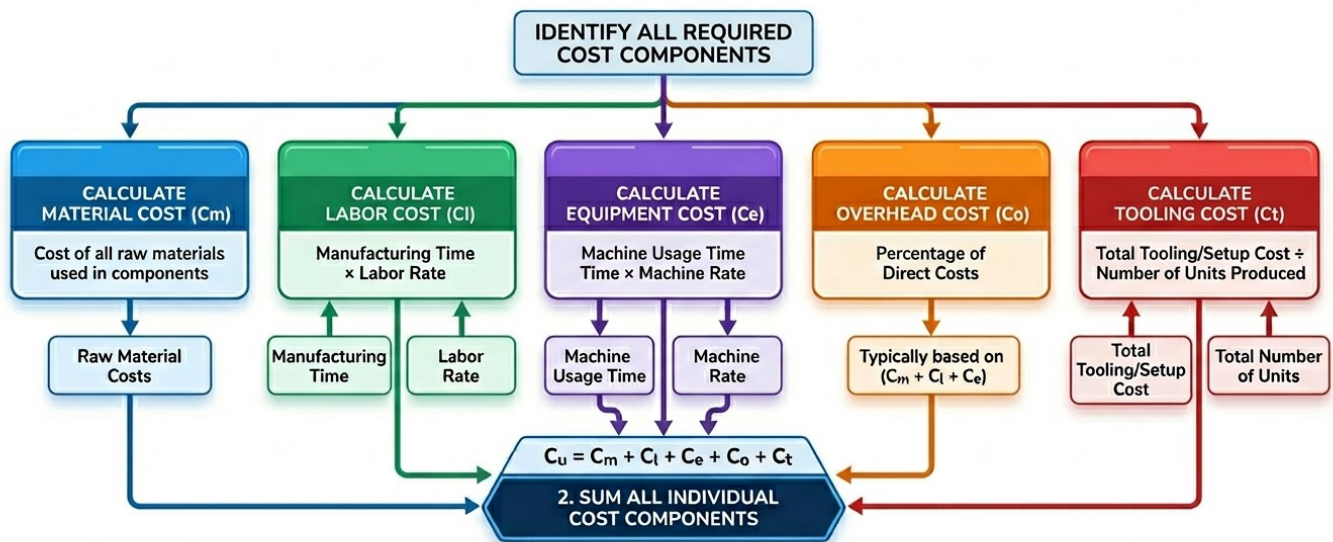


Figure 26. Cost Analysis Methodology.

4.4.1. Material Cost Estimation

The material cost represents the largest portion of the total unit cost and is estimated based on the quantity of raw materials required for each major component, along with typical market prices for the selected materials. The design primarily utilizes abrasion-resistant and structural steels to ensure durability under harsh operating conditions, particularly for handling crushed limestone. Components such as the screw flights are made from AR400 steel to resist wear, while structural elements like the shaft and trough use standard carbon and alloy steels for strength and manufacturability. In addition to fabricated components, several items such as bearings, seals, and the motor–gearbox assembly are treated as purchased components and included as lump-sum costs. The estimated material cost is summarized in Table 4.12.

Table 4.12. Material Cost Estimation

Component	Material	Weight/Quantity	Unit Cost(\$/lb)	Subtotal (\$)
Screw flights	AR400 steel	~850 lb	0.95	808
Central shaft / screw core	1045 steel	~600 lb	0.70	420
Trough body	Carbon steel	~900 lb	0.60	540
Covers, inlet, discharge, flanges	Carbon steel	~350 lb	0.60	210
End shafts / couplings	1045 steel	~250 lb	0.70	175
Hanger / support hardware	Carbon steel + hard iron	lump sum	—	180
Bearings and seals	Buy-out parts	lump sum	—	480
Motor + gearbox + drive coupling	Buy-out parts	1 set	—	2,400
Fasteners, guards, misc. hardware	Mixed	lump sum	—	300
Estimated material cost,				

The total estimated material cost is therefore \$5,513, with the motor and drive system contributing the largest single share. Among fabricated components, the trough body and screw flights represent significant cost contributors due to their material volume and the requirement for wear resistance. This cost breakdown provides a

realistic basis for further cost optimization and highlights key areas where material selection and design modifications could influence overall project economics.

4.4.2. Labor Cost Estimation

The labor cost is estimated based on the total manufacturing and assembly time required for the screw conveyor system, multiplied by typical industry labor rates for each operation. The fabrication process involves multiple stages, including plate preparation, flight forming, machining, welding, assembly, and inspection. Each stage requires different skill levels and equipment, which is reflected in the variation of hourly labor rates. Higher precision operations such as machining and welding contribute significantly to the overall labor cost due to their longer durations and higher skill requirements. The estimated labor cost breakdown is presented in Table 4.13.

Table 4.13. Labor Cost Estimation

Operation	Labor Time (hr)	Labor Rate (\$/hr)	Subtotal (\$)
Plate cutting or blank preparation	6	35	210
Flight forming or rolling	8	40	320
Machining of shaft, ends, seats, couplings	10	60	600
Welding and fabrication	14	45	630
Final assembly and alignment	8	35	280
Inspection or fit-up check	3	40	120
Estimated labor cost,			

The total estimated labor cost is therefore \$2,160, with welding, fabrication, and machining operations representing the largest contributions. This reflects the complexity of assembling a heavy-duty screw conveyor and the importance of maintaining dimensional accuracy, alignment, and structural integrity during fabrication.

4.4.3. Equipment Cost Estimation

The equipment cost accounts for the usage of manufacturing machinery required to fabricate and assemble the screw conveyor components. This cost is estimated based on the duration each machine is used and the corresponding hourly machine rate, which reflects operating expenses such as power consumption, maintenance, and depreciation. The fabrication process relies on a combination of cutting, forming, machining, and welding equipment, along with handling tools used during assembly and fit-up operations. More precise and capital-intensive machines such as CNC machining centers contribute a larger share of the equipment cost due to their higher hourly rates. The estimated equipment cost breakdown is presented in Table 4.14.

Table 4.14. Equipment Cost Estimation

Equipment / Machine	Estimated Time (hr)	Machine Rate (\$/hr)	Subtotal (\$)
Plasma or profile cutting machine	6	25	150
Rolling or forming equipment	8	30	240
Lathe or mill or CNC machining	10	50	500
Welding equipment	14	20	280
Handling and fit-up equipment	8	15	120
Estimated equipment cost,			

The total estimated equipment cost is therefore \$1,290, with machining operations contributing the highest portion due to their higher machine rates. This reflects the importance of precision machining in ensuring proper fit, alignment, and performance of rotating components within the screw conveyor system.

4.4.3. Overhead Cost Estimation

The overhead cost accounts for indirect expenses associated with manufacturing that are not directly tied to specific materials, labor, or machine usage. These include costs such as facility utilities, supervision, administrative support, maintenance, and general operational expenses. For a preliminary design and cost estimation, overhead is commonly approximated as a fixed percentage of the direct manufacturing cost, which includes material, labor, and equipment costs.

In this analysis, the overhead cost is assumed to be 20% of the total direct cost, providing a reasonable estimate for industrial manufacturing conditions. Based on the previously calculated values of material cost (), labor cost (), and equipment cost (), the overhead cost is determined as follows:

The estimated overhead cost is therefore approximately \$1,793, which reflects the additional indirect expenses required to support the manufacturing process. This value is consistent with typical early-stage cost models and provides a practical basis for evaluating the total production cost of the system.

4.4.4. Tooling Cost Estimation

The tooling cost for the screw conveyor system is estimated based on setup, fixtures, and jigs required for fabrication rather than high-cost mass production tooling. Since this is a low-volume industrial product, the cost is dominated by reusable setup and alignment fixtures used during machining, welding, and assembly operations. Therefore, tooling cost is reasonably modeled as an annual expense distributed across the expected production volume.

For this analysis, the total annual tooling and setup cost is assumed to be \$12,000, which includes expenses related to fixture fabrication, maintenance, and setup operations. With an estimated production volume of 250 units per year, the tooling cost per unit is calculated by evenly distributing this annual cost across all units produced.

The estimated tooling cost per unit is therefore \$48, which is relatively small compared to material and manufacturing costs. This reflects the low-volume nature of the product, where flexible and reusable tooling strategies are preferred over dedicated high-cost production tooling.

4.4.4. Total Estimated Cost Per Unit

The total manufacturing cost per unit is obtained by summing all individual cost components, including material, labor, equipment, overhead, and tooling costs. This comprehensive estimate provides a realistic assessment of the overall production expense for the screw conveyor system under the assumed operating and manufacturing conditions. Each cost component reflects a different aspect of the production process, from raw material procurement to fabrication, machining, assembly, and indirect operational expenses. The final cost breakdown is summarized in Table 4.15.

Table 4.15. Final Cost Estimation

Cost Element	Symbol	Value (\$)
Material cost		5,513
Labor cost		2,160
Equipment cost		1,290
Overhead cost		1,793
Tooling cost per unit		48
Total cost per unit		10,804

The total estimated manufacturing cost per unit is therefore \$10,804, with material cost being the dominant contributor, followed by labor and overhead costs. This result highlights the importance of material selection and

fabrication efficiency in controlling the overall cost of the system. The estimate serves as a baseline for further economic evaluation, optimization, and potential cost reduction strategies in the final design.

4.4.5. Cost Analysis Key Insights

The cost analysis reveals that material cost is the dominant contributor to the overall manufacturing cost, primarily due to the large quantity of steel required and the use of AR400 wear-resistant material for the screw flights. This highlights the importance of material selection, as durability requirements directly influence cost. While necessary for long service life in abrasive environments, these material choices significantly impact the total expense.

Labor cost also represents a substantial portion of the total cost, driven by fabrication-intensive processes such as welding, machining, and assembly. These operations require skilled labor and careful execution to ensure proper alignment and structural integrity, which increases both time and cost. Similarly, the equipment cost reflects the reliance on manufacturing processes rather than standardized off-the-shelf components, particularly for machining and forming operations.

In contrast, tooling cost per unit remains relatively low due to the low-volume production nature of the system. Instead of investing in expensive dedicated tooling, the design relies on reusable fixtures and setups, which are distributed across the production volume. The overhead cost, estimated as a percentage of direct manufacturing cost, remains moderate and consistent with typical preliminary cost models.

Overall, the total cost is largely influenced by the need for wear-resistant materials and robust welded construction, both of which are essential for reliable operation in harsh industrial conditions. This analysis suggests that any cost optimization efforts should focus on material efficiency and fabrication processes while maintaining the required performance and durability of the system.

4.5 Ergonomics and Safety Analysis

The ergonomics and safety analysis focuses on the final 30 in horizontal screw conveyor. The conveyor is designed for safe fabrication, operation, maintenance, cleaning, and inspection. The main hazards come from rotating parts, drive components, limestone dust, material spillage, electrical service, heavy components, and maintenance access. The design controls these hazards using covered moving parts, guarding, dust containment, overload protection, lockout/tagout procedures, warning labels, and accessible service points [2], [3].

The final design uses a covered trough so the rotating screw remains enclosed during operation. The drive end includes guards around the V-belt, coupling, shaft connection, and reducer. The inlet and discharge use formed transitions to reduce dust release and material leakage. The end seals and covers limit limestone dust movement toward bearings and nearby work areas. These features reduce direct contact with moving parts and help keep the surrounding area cleaner.

4.5.1. Design for Safety Approach

The safety approach follows four levels: eliminate hazards where the design permits it, guard remaining hazards, warn users with labels and signals, and use trained personnel with proper protective equipment. This approach is applied directly to the screw conveyor layout. The screw is placed inside a covered trough, exposed drive elements are guarded, warning labels are placed near hazard locations, and service tasks require lockout/tagout before work starts.

Table 4.16. Design for Safety Approach

Step	Design Action Used in Final Conveyor
Eliminate hazards	The screw is placed inside a covered trough, and service points are kept outside the material path where practical

Step	Design Action Used in Final Conveyor
Guard hazards	Fixed guards cover exposed shafts, coupling, V-belt drive, and rotating drive components
Warn users	Safety labels and indicator lights are placed near the drive, cover access, inlet, discharge, and control panel
Train and use PPE	Lockout/tagout, eye protection, hand protection, dust protection, and trained-personnel-only service are required

4.5.2. Main Safety Concerns and Design Controls

The highest safety concern is contact with moving components. The screw, shaft, coupling, V-belt, and reducer all create hazards during operation. For this reason, the final design keeps the screw enclosed and uses fixed guarding at the drive end. The guards remain in place during operation. Cover removal, cleaning, seal replacement, belt adjustment, and bearing service require lockout/tagout before work begins.

Table 4.17. Safety Concerns and Mechanisms Followed

Safety Concern	Design Control
Contact with rotating screw	Bolted covers enclose the trough during operation
Pinch points at drive components	Fixed guards cover the V-belt, coupling, reducer, and exposed shaft areas
Entanglement near shafts and couplings	Drive guards and warning labels block access during operation
Material jam or overload	Motor overload protection and controlled shutdown reduce damage risk
Limestone dust release	Covered trough, inlet transition, discharge spout, and seals reduce dust escape
Material spillage around conveyor	Dust-tight covers and controlled discharge reduce cleanup hazards
Electrical exposure during service	Control-panel access is limited to trained personnel, and lockout/tagout is required
Sharp edges on fabricated parts	Exposed edges and cover corners are deburred and rounded
Heavy component handling	Modular sections allow hoist or team lift handling during assembly and maintenance
Maintenance inside conveyor	Covers are removable only after shutdown and lockout/tagout

Dust is another major safety concern because the limestone is dusty and aerated. The conveyor uses a covered trough, shaft seals, inlet transition, and discharge spout to limit dust release. Less dust also improves ergonomics because operators spend less time cleaning material from the floor and nearby equipment. Spilled limestone creates slip and trip concerns, so the containment features also support a safer work area.

4.5.3. Ergonomic Improvements

The ergonomic layout focuses on inspection, cleaning, cover removal, bearing access, seal replacement, and drive service. These tasks occur during routine maintenance, so the final design keeps them accessible without forcing deep reaching into the conveyor body. External seals and end bearings improve service access. Modular conveyor sections reduce handling difficulty during fabrication and installation.

Table 4.18. Ergonomic Improvements and Rules Followed

Ergonomic Improvement	Result
Modular conveyor sections	Large parts are handled in smaller sections during fabrication, transport, and installation
External end bearings and seals	Inspection and replacement are easier because service points are outside the trough
Removable bolted covers	Operators inspect the screw and trough after shutdown and lockout
Guarded drive end	Belt, reducer, and coupling service is controlled and separated from normal operation

Ergonomic Improvement	Result
Rounded and deburred edges	Contact risk is reduced during assembly and maintenance
Control panel within normal reach	Operators start, stop, and monitor the conveyor without awkward movement
Indicator lights	Operating status is clear during startup, shutdown, and abnormal conditions
Hoist or team lift for heavy parts	Manual lifting strain is reduced for trough sections, screw sections, and drive components

The conveyor is not designed for operators to reach into the trough during operation. Inspection and cleaning are performed only after the machine is stopped, isolated, and verified safe. This matches the safety requirement for rotating machinery. The control panel is placed close enough for convenient use but outside the moving equipment area. Indicator lights give the operator clear status information during operation.

4.5.4. Safety Labels Applied

Safety labels as illustrated in Figure 27 are placed on the drive guard, cover access points, inlet, discharge, control panel, and lockout points. The labels identify rotating equipment, pinch points, required guards, lockout/tagout, PPE, trained-personnel-only service, and dust exposure precautions. These labels support safe use by reminding operators and maintenance workers of the hazards at the exact locations where they occur. The final ergonomics and safety design supports the project goals because it reduces maintenance difficulty, controls dust and spillage, improves service access, and protects operators from rotating parts. The covered trough, guarded drive, modular construction, accessible seals, external bearings, and safety labels make the final conveyor safer and easier to maintain in a limestone handling environment [4].



Figure 27. Safety Labels.

4.6 CAD Design and Modeling

The selected concept was further developed into a detailed CAD model to define the geometry, assembly structure, and component-level design of the screw conveyor system. The CAD modeling phase focused on creating a modular and manufacturable design while maintaining compatibility with standard industrial components. The complete assembly includes the screw conveying system, trough and cover system, drive unit, and supporting components such as hangers and bearings. The design was developed as a sectional assembly to allow ease of manufacturing, transportation, and maintenance.

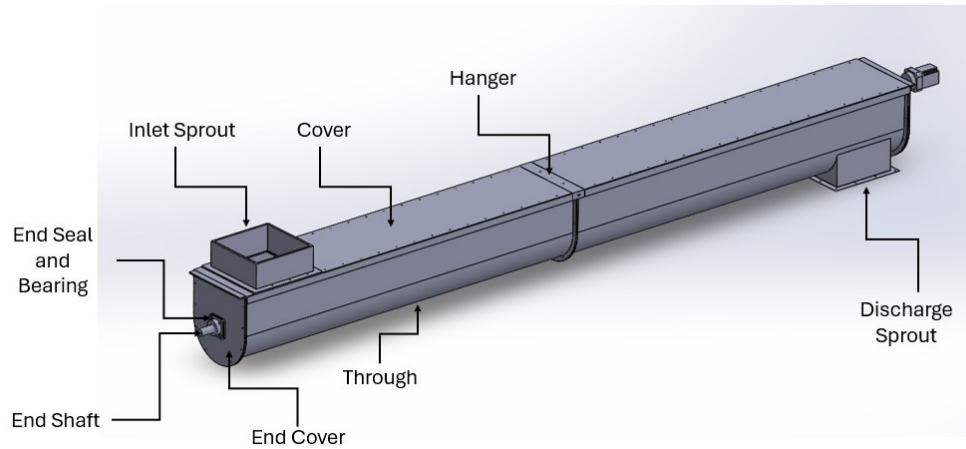


Figure 28. Complete Screw Conveyor Assembly.

4.6.1. Assembly Overview

The screw conveyor assembly consists of multiple integrated subsystems that work together to ensure continuous material transport. The inlet spout introduces material into the system, while the rotating screw conveys it along the trough to the discharge spout. The trough serves as the primary containment structure, while covers ensure dust-tight operation. Hangers are strategically placed to support the screw shaft along its length, preventing excessive deflection and maintaining alignment. The drive system, consisting of a motor and gearbox, provides the required torque to rotate the screw. The design adopts a sectional configuration, where the screw and trough are divided into manageable segments. This approach simplifies fabrication, assembly, and maintenance operations.

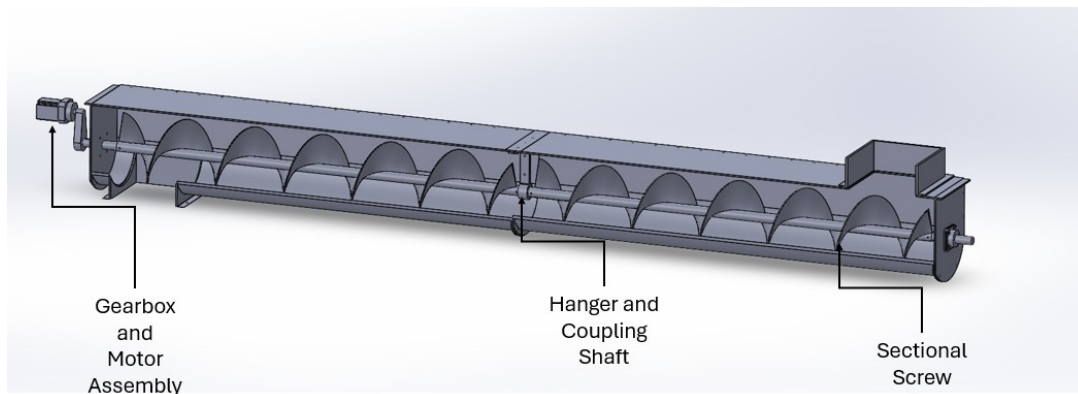


Figure 29. Sectional View of Screw Conveyor Assembly.

4.6.2. Functional Component Description

Each component in the assembly was designed with a specific functional role, ensuring efficient system operation. Critical components such as the screw, shafts, bearings, and hanger supports directly influence the performance and reliability of the system.

Table 4.18. Screw Conveyor Component Function Table

Component	Function	Criticality
Sectional Screw	Rotates to convey bulk material along the trough	Critical
Trough	Contains and guides the material flow during conveying	Critical

Cover	Encloses the trough to prevent spillage and contamination	--
Cover with Inlet	Provides controlled entry point for material into the conveyor	--
Inlet Spout	Directs incoming material into the screw conveyor	--
Discharge Spout	Allows controlled exit of material from the conveyor	--
Hanger	Supports the screw shaft mid-span to prevent deflection and misalignment	Critical
Coupling Shaft	Connects screw sections and transmits torque between them	Critical
Drive Shaft	Transfers torque from the motor/gearbox to the screw	Critical
End Shaft	Supports and aligns the screw at the non-driven end	Critical
End Bearing	Supports the rotating shaft and reduces friction at the ends	Critical
End Cover	Closes the trough end and supports bearing/shaft alignment	--
End Seal	Prevents leakage of material and ingress of contaminants around the shaft	Critical
Gearbox and Motor Assembly	Provides controlled rotational power to drive the screw	--

The screw (flights and shaft) is responsible for material transport and is considered the most critical component. The trough ensures containment and guides material flow, while the hanger supports minimize shaft deflection. The drive shaft and coupling shaft transmit torque from the motor to the screw assembly. Non-critical components such as covers, inlet, and discharge spouts primarily contribute to containment and operational convenience.

4.7 Component-Level CAD Design

Detailed CAD models were developed for each individual component to ensure manufacturability and proper assembly fit. The design follows standard mechanical design practices, including the use of uniform thicknesses, standard shaft geometries, and compatible tolerances.

The sectional screw was modeled to allow modular assembly, enabling ease of replacement and reduced manufacturing complexity. The trough and cover components were designed using sheet metal features, making them suitable for forming processes. Shafts and bearings were modeled with standard dimensions to ensure compatibility with commercially available components.

Special attention was given to the hanger assembly, which plays a critical role in maintaining shaft alignment and minimizing deflection. The coupling shaft design allows torque transmission between screw sections while accommodating assembly tolerances.

4.7.1. Design Considerations for Manufacturability

The CAD design was developed with manufacturability considerations in mind. Sheet metal components such as the trough and covers were designed with standard bend angles and uniform thickness to facilitate forming operations. Machined components such as shafts and hanger parts were designed to minimize complex geometries and allow easy fixturing during machining.

The use of modular sections reduces fabrication complexity and enables parallel manufacturing of components. Standardization of parts, such as bearings and fasteners, further reduces cost and simplifies procurement.

Additionally, the design minimizes unnecessary geometric complexity, ensuring that most components can be manufactured using conventional processes such as turning, milling, and sheet metal forming.

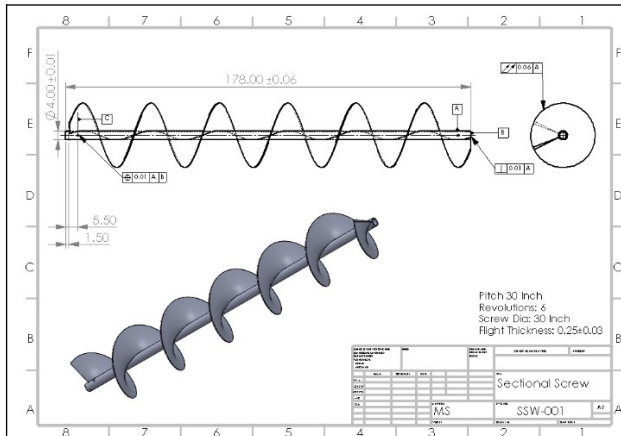


Figure 30. Sectional Screw CAD Model.

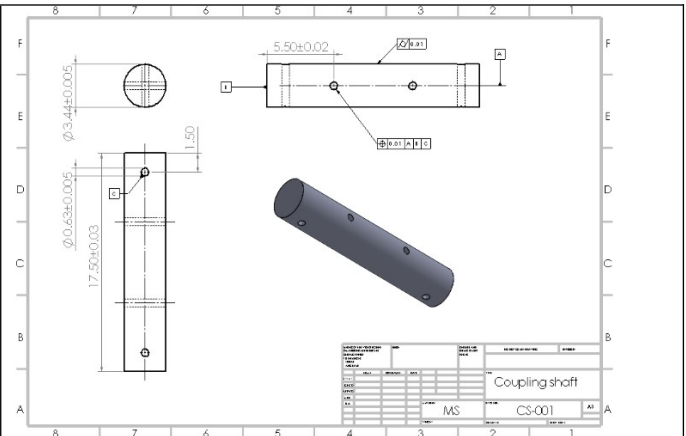


Figure 31. Coupling Shaft CAD Model.

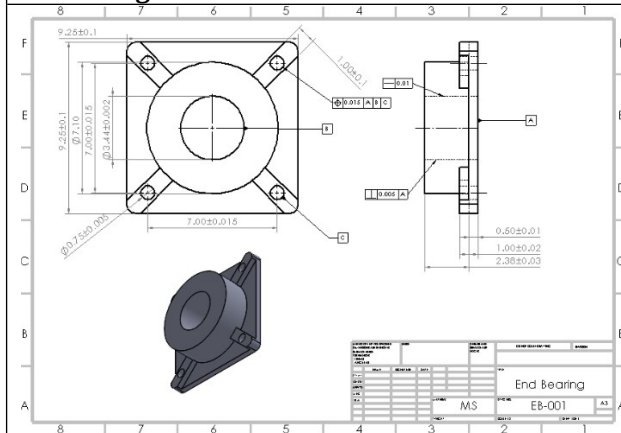


Figure 32. End Bearing CAD Model.

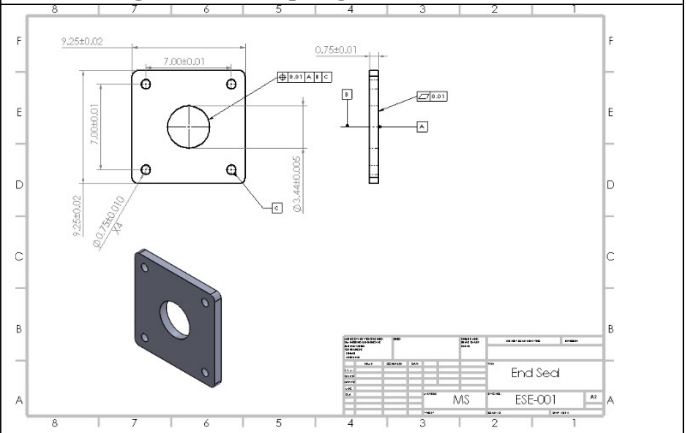


Figure 33. End Seal CAD Model.

4.7.2. Design Considerations for Assembly

From an assembly perspective, the CAD design incorporates features that facilitate alignment and ease of installation. The sectional screw design allows assembly in stages, reducing handling difficulty. The use of couplings enables easier alignment between screw segments.

Hangers are positioned to provide adequate support while maintaining accessibility for installation and maintenance. However, the design currently relies on a large number of fasteners, which increases assembly time and complexity, as identified in the DFA analysis.

Future improvements may include reducing fastener count and integrating components where it is possible to simplify assembly operations.

5. Conclusion

This project successfully developed a complete and practical design for an industrial screw conveyor system intended to handle crushed limestone under realistic operating conditions. A structured engineering design approach was followed, beginning with translating customer requirements into engineering characteristics using the House of Quality, and progressing through concept generation, parametric design, material selection, manufacturing planning, and final evaluation. The final design meets the required capacity of 150 TPH over a 30 ft horizontal distance while maintaining reliable operation, effective dust containment, and compatibility with standard industrial components.

A key outcome of the design process is the selection of a 30-inch screw conveyor operating at approximately 15.9 RPM, which provides a well-balanced solution between performance, cost, and maintenance requirements. The lower operating speed reduces wear, dust generation, and power demand, while the reduction of hanger bearings minimizes internal obstruction and improves flow stability. The material selection strategy further supports durability, particularly through the use of AR400 steel for screw flights, which significantly enhances wear resistance in abrasive limestone conditions. Overall, the design demonstrates that a larger, slower screw configuration is more effective for long-term reliability compared to smaller, high-speed alternatives.

From a manufacturing and assembly perspective, the design aligns well with Design for Manufacturing principles by utilizing fabrication-based processes such as welding, machining, and sheet metal forming, which are appropriate for low-volume industrial production. However, the Design for Assembly analysis highlights that the system has relatively high assembly complexity due to the large number of components and fasteners. This indicates that while the design is manufacturable, there are inefficiencies in assembly that could increase labor time and cost.

The analytical evaluations confirm that the system performs reliably under expected conditions. The overall system reliability is approximately 97.11%, indicating strong performance for an industrial conveying system operating in a dusty and abrasive environment. The safety factor analysis shows that all critical components operate within acceptable limits, ensuring safe and stable operation. However, the reliability analysis also identifies hanger bearings and drive components as the most critical elements, as they are more susceptible to wear, misalignment, and maintenance-related issues.

Despite the strengths of the design, several limitations are evident. The system still relies on hanger bearings, which remain the most failure-prone components due to their exposure to abrasive material and dust. The high number of fasteners contributes to assembly complexity and may increase maintenance effort. Additionally, the use of wear-resistant materials such as AR400, while necessary for durability, increases material cost and affects overall economic efficiency. The analysis is also primarily based on steady-state conditions and does not fully account for dynamic effects, transient loading, or long-term fatigue behavior under variable operating conditions.

Future improvements can focus on enhancing both performance and manufacturability. Reducing the number of fasteners and integrating components where possible would improve assembly efficiency and reduce labor costs. The introduction of modular subassemblies could simplify installation and maintenance while improving system reliability. Further improvements in hanger bearing design, including better sealing and lubrication systems, could significantly increase system reliability. Exploring alternative materials or surface treatments may provide improved wear resistance at lower cost. Additionally, incorporating real-time monitoring systems such as vibration or load sensors would enable predictive maintenance and reduce unexpected failures. Finally, extending the analysis to include dynamic and fatigue behavior would provide a more comprehensive understanding of long-term performance.

Overall, the final design represents a balanced and practical engineering solution that satisfies the project objectives while clearly demonstrating the trade-offs between performance, cost, manufacturability, and maintenance.

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